

# ChoCo: a Chord Corpus and a Data Transformation Workflow for Musical Harmony Knowledge Graphs

Jacopo de Berardinis<sup>1,\*</sup>, Albert Meroño-Peñuela<sup>1</sup>, Andrea Poltronieri<sup>2,\*</sup>, and Valentina Presutti<sup>3</sup>

<sup>1</sup>Department of Informatics, King's College London, 30 Aldwych, London WC2B 4BG, UK

<sup>2</sup>Department of Computer Science and Engineering, University of Bologna, Mura Anteo Zamboni, 7, Bologna 40126, IT

<sup>3</sup>LILEC, University of Bologna, Via Cartoleria, 5, Bologna 40124, IT

\*corresponding author(s): Jacopo de Berardinis (jacopo.deberardinis@kcl.ac.uk), Andrea Poltronieri (andrea.poltronieri2@unibo.it)

## ABSTRACT

Various disconnected chord datasets are currently available for music analysis and information retrieval, but these are limited by either their size, non-openness, lack of timed information, and interoperability. Together with the lack of overlapping repertoire coverage, this limits cross-corpus studies on harmony over time and across genres, while hampering research in computational music analysis (chord recognition, pattern mining, computational creativity), where machine learning methods are notoriously eager for data. To address this gap, we propose the Chord Corpus (ChoCo), a large-scale dataset that semantically integrates harmonic data from 18 different sources in various representations and formats (Harte, Leadsheet, Roman numerals, ABC, etc.). We leverage JAMS (JSON Annotated Music Specification), a popular data structure for annotations in Music Information Retrieval, to effectively represent and enrich a variety of chord-related information (chord, key, mode, etc.) in a uniform way. On top of it, we propose a novel ontology modelling music annotations and involved entities (artists, scores, etc.), build a 30M triple knowledge graph, and find 4K+ links to other datasets (MIDI-LD, LED).

## 1 Background & Summary

Western tonal music encompasses several dimensions (melody, harmony, rhythm, etc.) and temporal scales (beat, measure, phrase, etc.), which all contribute to characterise a complex signal studied in different fields and from various lenses. One prominent dimension is represented by harmony, also known as the “vertical dimension” of music, which is concerned with “combining notes in music to produce a pleasing effect greater than the sum of its parts”<sup>1</sup>. The study of harmony has a long history. Aristoxenus authored “Elements of Harmony”, which is considered to be the first work on harmony in European history. The study of harmony continued centuries later, when in the 18<sup>th</sup> century Euler proposed a grading system of chord aesthetics, which aimed to establish a hierarchy of musical consonance and dissonance. Since then, several esteemed mathematicians, physicists, and philosophers have contributed to the description and formalisation of harmonic concepts<sup>2</sup>.

Chords are the basic constituents of harmony, which jointly define the harmonic structure of a piece. Individually, a chord is defined as a simultaneous occurrence of several music sounds, producing harmony<sup>3</sup>. Depending on the notational system and the annotation conventions, a chord can be associated with a name, or label. For example, in the Leadsheet notation, B $\flat$ <sup>7</sup>(#9) (typically read as “B flat dominant seventh sharp nine”) consists of 5 simultaneous notes: B $\flat$ , D, F, A $\flat$ , and D $\flat$ , and in western music is usually associated with a startling, dissonant, and irresolute sound. The distances between notes (or intervals) in this chord are the root (B $\flat$ ), its major third, its fifth, its seventh, and its augmented ninth. These intervals to the root characterise the intrinsic harmonic properties of chords, as well as the relationships with other chords in the same harmonic progression<sup>4</sup>.

In the broader context of music, harmony is a widely studied component in theory<sup>4,5</sup>, and is also of particular interest for music analysis<sup>6</sup>. In tonal music, functional harmony provides a set of rules for moving to and from the *tonic* – the most stable note in a piece, in an orderly fashion, allowing to relate chords to each other, and to the main harmony. In Yehudi Menuhin’s words, “music creates order out of chaos: for rhythm imposes unanimity upon the divergent, melody imposes continuity upon the disjointed, and harmony imposes compatibility upon the incongruous.” Although what defines congruence can be ascribed to a musical theory, these functional rules and relations can also be traced to psychological cues.

Perceptually, some chords sound more stable, final and resolved, while others sound unstable and tense – a phenomenon that is salient both to young children and to adults, even from diverse cultures. However, the definition of harmony differs vastly across time, genre, and individuals<sup>7</sup>, reflecting a great heterogeneity in terms of harmony perception<sup>8,9</sup> (here we focus on

Western tonal music tradition). In this regard, harmony exerts an affective role: major harmonies tend to represent positive emotions (happiness, joy, triumph, etc.); minor triads express “negative” emotions (sadness, anger, etc.); diminished triads (chains of minor thirds) indicate suspense and other disorienting sentiments, while augmented triads (all major third intervals) tend to create senses of spookiness, extreme dark emotions, and mystery<sup>1</sup>.

Computationally, the automatic analysis of chord progressions has allowed to address several tasks in information retrieval – from the detection of cadences, patterns, structures in music, to the introduction of harmonic similarity measures for cover song detection, symbolic search, and content-based music linking. Progress in machine learning research has also sparked interest in computational creativity applications, such as arrangement generation, continuation, infilling, and automatic music composition with harmonic conditioning<sup>10</sup> (e.g. generating melodies from a given harmonic template) to name a few.

To account for the evolution of harmony and explain its subjective and genre-specific differences, while enabling the aforementioned applications, the availability of large, diverse, and reliable chord data is fundamental. However, several different chord notations exist (Harte, Roman, ABC, Leadsheet, etc.), each with different levels of expressiveness, in a large number of disconnected chord datasets that are hard to combine<sup>11</sup>. This poses a challenge for combining existing chord datasets into larger ones. Other existing approaches address this by typically focusing on scale, and publish large numbers of chord annotations. For example, UltimateGuitar (<https://www.ultimate-guitar.com/>) offers a collection of 1.1M+ songs annotated by a community of 12M+ musicians. Chordify addresses the challenge of scalable chord annotation by applying methods for automated chord estimation. However, none of these approaches solves the problem of integrating chord annotation datasets with the following desiderata: **(a)** sufficiently high quality for downstream tasks; **(b)** precise timing information; **(c)** released through open licenses; **(d)** using different chord notations; **(e)** diversity of music genres; and **(f)** at large scale. The problem is exacerbated by the lack of a standard music annotation format. Here, a music annotation is defined in a broader sense as the outcome of a music analysis carried out by a domain expert on the musical surface (a score, a recording) to identify and locate elements of interest (e.g. chords, beats, segments, patterns, etc.), following a well-established methodology. For example, if the goal of a harmonic analysis is to identify chords from a composition, a music annotation may correspond to a list of chords together with a reference to their onset and offset (i.e. when they occur in the piece).

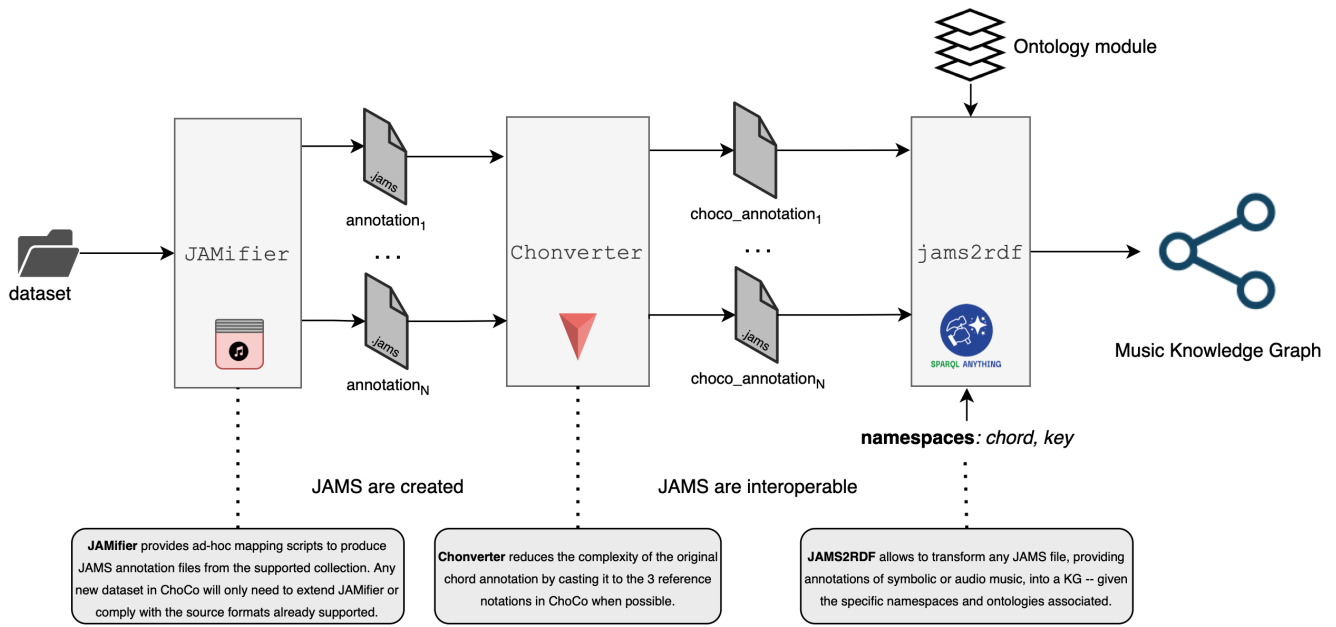
## 1.1 The problem of music data scarcity and interoperability

In the last decade, numerous systems and formats have been proposed for representing and storing musical annotations<sup>12</sup>. While some have been more successful than others, no single system has prevailed as the main standard. Some systems are focused on symbolic music and are domain-specific (e.g. DCMLab, RomanText for harmonic analyses), embed annotations in the score (MusicXML, ABC, etc.), or propose variations of tabular formats to account for audio and symbolic music (LAB and xLAB). In the audio domain, JAMS (JSON Annotated Music Specification)<sup>13</sup> has emerged as a system to uniformly represent music annotations of different type and granularity, that is efficiently built on top of the JSON serialisation standard. Notably, JAMS is also supported by software libraries for dataset manipulation<sup>14</sup> and for the evaluation of MIR methods<sup>15</sup>.

However, combined efforts of MIR and Semantic Web (SW) researchers to address (chord) annotation data interoperability have been scarce. While MIR has contributed a great deal of music datasets, predominantly containing music annotations to train and evaluate computational methods for music analysis, SW technologies and principles can easily address the data integration problem at scale<sup>11</sup>. Nevertheless, the scarcity of semantic models for music annotations has hampered this vision, and more research efforts are hence necessary to devise domain-specific ontologies that can efficaciously address the interoperability issue through reuse and alignment. In addition, this kind of musical knowledge is also underrepresented in Knowledge Graphs<sup>16</sup>, which are typically built from other knowledge archetypes such as logic statements or textual corpora. The lack of musical knowledge in the Semantic Web also limits our understanding of knowledge expressed in modalities other than text (e.g. images, music) and its challenges: semantic relations that have not been formalised yet, integration of multimodal datasets, etc.

Specifically for harmonic data, various chord collections have been previously published (see Table 1) making harmony annotations available, albeit through highly heterogeneous and non-interoperable notations (Harte, Leadsheet, Roman, ABC) and formats (JAMS, JSON, MusicXML, LAB, etc.). Other databases, such as UltimateGuitar and Chordify<sup>17</sup>, focus on automation and scalability. These are achieved by annotating millions of songs via crowdsourcing or chord recognition algorithms, but have an inherent cost in annotation quality. Therefore, none of these approaches solves the problem of semantically integrating chord annotation datasets while meeting all the aforementioned desiderata (a-f).

The challenge of supporting interoperability of music content-related data has been the subject of relevant efforts in the last decade, especially supporting their evolution, reuse, and sustainability<sup>18–20</sup> according to FAIR data principles<sup>21</sup> and through Semantic Web technologies. The Music Ontology<sup>22</sup> addresses contextual metadata about music pieces, such as when they were recorded or arranged and by whom, providing a basis for interlinking music datasets. The OMRAS2 Chord Ontology<sup>23</sup> defines a vocabulary to describe chords and chord sequences in RDF, albeit is no longer maintained and does not support reasoning. The HaMSE Ontology<sup>24</sup> aligns different music representation systems and describes a set of musicological features to enable different degrees of interoperability. MusicOWL<sup>25</sup> describes the structure of music scores to support MIR tasks



**Figure 1.** Overview of our data transformation workflow, generalised for arbitrary music annotations, and used here for chord and key annotations prior to the generation of the ChoCo Knowledge Graph. The `JAMifier` ingests chord collections (where metadata and music annotations follow collection-specific conventions and formats) to generate a JAMS dataset. This achieves two integration levels, as all *metadata* are consistently re-organised, and the *music annotations* (i.e. chord progressions, in this case) are all encoded and stored in separate JAMS files – one per track/score. The `Chonverter` achieves notational interoperability among collections by converting the original annotations to the same notational families. Finally, `jams2rdf` leverages notation-specific ontologies to generate RDF triples and create a Music Knowledge Graph.

using musical features rather than text. It addresses concepts such as clef, dynamic, key, note, etc., and reuses the Tonality Ontology (<http://motools.sourceforge.net/doc/tonality.html>) on top of the Music and Chord ontologies. The Music Theory Ontology (MTO)<sup>26</sup> adds music theoretical notions that were overlooked in existing ontologies or only partly defined, such as musical notation, duration, progression and degree, aligning with e.g. the Chord Ontology and supporting deductive inferences such as scale degree and chord degree. The Diatonic-Chromatic System Ontology<sup>27</sup> uses reasoning to infer if a score can be classified within the analytical framework of Michael Praetorius (1571–1621). A lightweight RDF vocabulary for representing events contained in MIDI files has been proposed with the MIDI Ontology<sup>28</sup>.

Some of these ontologies are the backbone of large music notation knowledge graphs. For example, the MIDI Linked Data Cloud<sup>28</sup> has been used to generate the MIDI Linked Data Cloud (<https://midi-ld.github.io/>), a large knowledge graph interconnecting 300K+ MIDI files through 10B+ triples of music-related linked data addressing music content rather than metadata. This misses, however, explicit chord information that could be useful for the symbolic analysis of harmony. MusicOWL<sup>25</sup> has been used for producing the Linked Music Score Dataset (<https://linkeddata.uni-muenster.de/datasets/opendata/ulb/musicscores/>) knowledge graph, representing elements of 43 historical scores from the Münster University Library. Yet, none of these previous efforts successfully addresses the aforementioned challenges (issues a-e); especially providing representations that meet the standards and the needs of different communities (e.g. JAMS for MIR, Musicology, and RDF for Semantic Web, Digital Humanities, etc).

## 1.2 Our contribution

We present here the **Chord Corpus (ChoCo)**<sup>29</sup>, a large dataset for musical harmony knowledge graphs. We describe the data workflow to curate, transform, and integrate more than 20,000 human-made, high-quality harmonic annotations from 18 highly heterogeneous chord datasets (desiderata *a, b, f*), following the JAMS data structure as annotation model. The resulting annotations are rich in provenance data (e.g. metadata of the annotated work, authors of annotations, identifiers, etc.) and refer to both symbolic music notation and audio recordings, while encompassing different notation systems (desideratum *d*). After semantically enriching, extending, and standardising these annotations under the JAMS definition, we leverage our ontologies to release the ChoCo Knowledge Graph – providing fine-grained semantic descriptions of chords, opportunities for chord interoperability, and 3,000+ links to external datasets. All data and code are released using open data licences (desideratum *c*).

| Collection           | Type | Notation  | Original format | Annotations      | Genres    | Ref                |
|----------------------|------|-----------|-----------------|------------------|-----------|--------------------|
| Isophonics           | A    | Harte     | LAB             | 300              | pop, rock | <a href="#">31</a> |
| JAAH                 | A    | Harte     | JSON            | 113              | jazz      | <a href="#">32</a> |
| Schubert-Winterreise | A, S | Harte     | csv             | 25 (S), 25*9 (A) | classical | <a href="#">33</a> |
| Billboard            | A    | Harte     | LAB, txt        | 890 (740)        | pop       | <a href="#">34</a> |
| Chordify             | A    | Harte     | JAMS            | 50*4             | pop       | <a href="#">7</a>  |
| Robbie Williams      | A    | Harte     | LAB, txt        | 61               | pop       | <a href="#">35</a> |
| The Real Book        | S    | Harte     | LAB             | 2486             | jazz      | <a href="#">36</a> |
| Uspop 2002           | A    | Harte     | LAB             | 195              | pop       | <a href="#">37</a> |
| RWC-Pop              | A    | Harte     | LAB             | 100              | pop       | <a href="#">38</a> |
| Weimar Jazz Database | A    | Leadsheet | SQL             | 456              | jazz      | <a href="#">39</a> |
| Wikifonia            | S    | Leadsheet | mxl             | 6500+            | various   | -                  |
| iReal Pro            | S    | Leadsheet | iReal           | 2000+            | various   | -                  |
| Band-in-a-Box        | S    | Leadsheet | mgu, sku        | 5000+            | various   | <a href="#">40</a> |
| When in Rome         | S    | Roman     | RomanText       | 450              | classical | <a href="#">41</a> |
| Rock Corpus          | S    | Roman     | har             | 200              | rock      | <a href="#">42</a> |
| Mozart Piano Sonata  | S    | Roman     | DCMLab          | 54 (18)          | classical | <a href="#">6</a>  |
| Jazz Corpus          | S    | Hybrid    | txt             | 76               | jazz      | <a href="#">43</a> |
| Nottingham           | S    | ABC       | ABC             | 1000+            | folk      | <a href="#">44</a> |

**Table 1.** Overview of the 18 chord datasets currently included in ChoCo. Letters “A” and “S” are used to denote *audio* and *symbolic* (or *score*) music partitions, respectively – from which harmonic annotations are collected.

We also show evidence of interest and use of ChoCo, and postulate its value for the Semantic Web and MIR communities at enabling the study of harmony through large scale data. Specifically, the main contributions are summarised as follows.

- A large dataset and knowledge graph standardising, enriching, and integrating 18 existing chord collections in the literature. ChoCo is released both as a JAMS dataset and an RDF knowledge graph, to accommodate the requirements and needs of different communities (Music Information Retrieval, Musicology, Semantic Web, etc.).
- A generalised data curation framework to semantically integrate MIR harmonic datasets and represent chords from a large variety of formats (JSON, CSV, LAB, TXT, SQL, MusicXML, iReal, mgu, sku, ABC, etc.) as JAMS annotations.
- An ontological and extensible model to represent JAMS annotations, and chord annotations in particular, as RDF knowledge graphs, together with a set of SPARQL queries to extract chord-related information from JAMS files directly or transform them into RDF, using state of the art data engineering methods.

Notably, ChoCo achieves interoperability of harmonic datasets at three different levels: metadata, annotation format, and chord notation. The former two operations are implemented by integrating metadata from different sources, at the parsing level, and leveraging the JAMS annotation standard to store harmonic annotations consistently. The latter is achieved by converting chords to three reference notational systems (desideratum *d*) – bridging them via the Harte notation<sup>30</sup>. Together, these operations allow to use the integrated collections as if they belong to the same dataset, which in turn, enables to the automatic generation of Musical Knowledge Graphs. Nevertheless, in addition to the conversions, ChoCo provides the original annotations in the JAMS files, along with rich provenance descriptions that allow to keep track of the original source of each harmonic annotation.

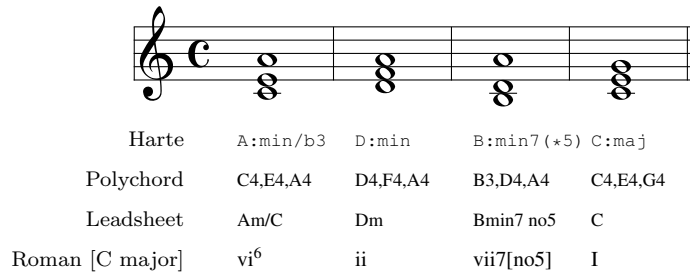
## 2 Methods

The general workflow is illustrated in Figure 1 and detailed in the following subsections. We describe here the resources contained in ChoCo (Section 2.1), and the data transformation workflow to: produce JAMS datasets with the *Jamifier* (Section 2.2), integrate the different chord notations via the *Chonverter* (Section 2.2.2), and create knowledge graphs by leveraging our JAMS ontology (Section 2.3).

### 2.1 Chordal data in ChoCo

Table 1 summarises the source chord datasets (alias *subsets*, *collections*) that are integrated in our framework. ChoCo v1.0 integrates 18 high-quality chord datasets providing timed annotations of chord progressions in different formats (e.g. LAB, CSV, txt, mxl), notations (e.g. Harte, Leadsheet, Roman, ABC), and types (audio, symbolic). The rich and diversified nature of

144 this resource, encompassing several genres/styles and periods, makes it the largest chord collection of its kind – with more than  
 145 20K annotated progressions. ChoCo’s collections can be categorised according to their generalised chord notation system:  
 146 Harte, Polychord, Leadsheet, and Roman. An example of notation systems for the same chord progression is given in Figure 2.



**Figure 2.** Example of a harmonic progression annotated using different notation systems, namely (i) Harte, (ii) Polychord (or decomposed chords), (iii) Leadsheet, and (iv) Roman Numerals.

147 **Harte collections** – gathering all collections with chords expressed in Harte notation<sup>30</sup>. The majority of these datasets are  
 148 focused on pop/rock music, released in LAB format, and collected from audio music. Among them, *Isophonics*<sup>31</sup> provides  
 149 chord, key, and structural annotations of a selection of albums by The Beatles, Queen, Michael Jackson, and Carole King;  
 150 *Billboard*<sup>34</sup> contributes similar annotations for a collection of songs sampled from the Billboard “Hot 100” chart in the United  
 151 States between 1958 and 1991; *Chordify Annotator Subjectivity Dataset (CASD)*<sup>7</sup> augments a subset of Billboard with 4 expert  
 152 annotators per song – to demonstrate the highly subjective nature of the chord identification/labelling task; *Robbie Williams*<sup>35</sup>  
 153 contains key and chord annotations for 5 albums from this artist; *Uspop2002*<sup>37</sup> is a large scale dataset for music similarity,  
 154 providing audio features, style tags, artist similarity ratings, as well as harmonic annotations for a smaller subset; *RWC-Pop*  
 155 is a subset of the the Real World Computing (RWC) database<sup>38</sup>, a cornerstone collection in MIR containing a great deal of  
 156 instrumental and performance annotations, in addition to chordal information that was contributed by LabROSA. Among  
 157 the other (non-pop) collections, we find the *Real Book*<sup>36</sup>, providing chord annotations of several jazz standards from the  
 158 homonymous book<sup>45</sup>; the *Audio-Aligned Jazz Harmony (JAAH)* dataset<sup>32</sup> contributing time-aligned harmony transcriptions  
 159 from “The Smithsonian Collection of Classic Jazz” and “Jazz: The Smithsonian Anthology”; and finally, the *Schubert*  
 160 *Winterreise*<sup>33</sup> multi-modal dataset, containing harmony and segment information of Franz Schubert’s song cycle “*Winterreise*”  
 161 which were separately annotated from the score and from the audio (9 performances per score).

162 **Leadsheet collections** – comprising four ChoCo collections using different flavours of the Leadsheet notation<sup>46</sup> for a  
 163 variety of genres. These include the *Weimar Jazz Database*<sup>39</sup>, providing rich cataloguing information, scores, YouTube links,  
 164 and harmonic/melodic annotations of a selection of jazz solo transcriptions; *Wikifonia*, a copyright-free online publisher of  
 165 sheet music in MusicXML format which was discontinued in 2013; the *Band-in-a-Box (BiaB) Internet corpus*<sup>40</sup>, containing  
 166 human-generated chord annotations for BiaB – a commercial software (<https://www.pgmusic.com>) that is used to  
 167 generate accompaniment for musical practice; the *iReal pro* collection, a newly contributed chord dataset of various genres (jazz,  
 168 blues, brazilian, latin, country, pop) that was created from the public playlists of iReal Pro (<https://www.irealpro.com>)  
 169 – a commercial app with similar functionalities to BiaB.

170 **Roman collections** – containing chord datasets providing harmonic annotations in Roman notation<sup>47</sup>, and with more  
 171 emphasis on classical music. A central dataset here is *When in Rome*<sup>41</sup>, which already contains harmonic analyses from the  
 172 *TAVERN* collection<sup>48</sup> (theme and variations for piano by Mozart and Beethoven), and the *BPS-FH* dataset<sup>49</sup> (Beethoven piano  
 173 sonata); but also harmonic annotations from Monteverdi madrigals, Bach chorales and preludes, Haydn Op. 20 String Quartets,  
 174 and a subset of nineteenth-century songs from the OpenScore Lieder corpus (*Winterreise* and *Schwanengesang* cycles from  
 175 Schubert, *Dichterliebe* from Schumann, and several pieces by female composers). Notably, *When in Rome* is an actively  
 176 maintained corpus where new harmonic annotations (in RomanText format) are also contributed and internally validated by  
 177 experts. As a growing corpus of functional harmonic analyses, we plan to support the integration of future releases within  
 178 ChoCo. Other Roman collections include the *Rock Corpus*<sup>42</sup>, providing harmonic analyses, melodic transcriptions and lyrics  
 179 information produced from a sample of Rolling Stone magazine’s list of the “500 Greatest Songs of All Time” in 2004 (pages  
 180 65 - 165); and *Mozart Piano Sonata*<sup>6</sup>, featuring harmonic, phrase, and cadence analyses of all piano sonatas by Mozart.

181 **Other collections** include *Nottingham*<sup>44</sup>, a dataset of British and American folk tunes, (hornpipe, jigs, etc.) released in ABC  
 182 format; and the *Jazz Corpus*<sup>43</sup>, providing harmonic analyses of jazz standards using both Harte-like and functional notations,  
 183 the latter of which is akin, in purpose, to Roman numerals.



| Resource                         | Link                                                                                                                                                    |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| ChoCo dataset                    | <a href="http://w3id.org/polifonia/resource/choco/">http://w3id.org/polifonia/resource/choco/</a>                                                       |
| Portal page                      | <a href="https://smashub.github.io">https://smashub.github.io</a>                                                                                       |
| JAMS Vocabulary namespace        | <a href="http://w3id.org/polifonia/ontology/choco/">http://w3id.org/polifonia/ontology/choco/</a> (prefix jams)                                         |
| JAMS Resource namespace          | <a href="http://w3id.org/polifonia/resource/choco/">http://w3id.org/polifonia/resource/choco/</a> (prefix pon-res)                                      |
| Roman Chord Vocabulary namespace | <a href="http://w3id.org/polifonia/ontology/roman-chord/">http://w3id.org/polifonia/ontology/roman-chord/</a> (prefix roman)                            |
| GitHub organization & code       | <a href="https://github.com/smashub/">https://github.com/smashub/</a>                                                                                   |
| Dataset generation code          | <a href="https://github.com/smashub/choco">https://github.com/smashub/choco</a>                                                                         |
| Documentation and tutorials      | <a href="https://smashub.github.io/docs/category/choco-the-chord-corpus">https://smashub.github.io/docs/category/choco-the-chord-corpus</a>             |
| Example data story               | <a href="https://projects.dharc.unibo.it/melody/choco/chord_corpus_statistics">https://projects.dharc.unibo.it/melody/choco/chord_corpus_statistics</a> |
| VoID description                 | <a href="https://github.com/smashub/choco/blob/main/void.ttl">https://github.com/smashub/choco/blob/main/void.ttl</a>                                   |
| SPARQL endpoint                  | <a href="https://polifonia.disi.unibo.it/choco/sparql">https://polifonia.disi.unibo.it/choco/sparql</a>                                                 |
| Zenodo                           | <a href="https://zenodo.org/badge/latestdoi/462698362">https://zenodo.org/badge/latestdoi/462698362</a>                                                 |

**Table 2.** Links to key ChoCo resources: ontology, datasets, and knowledge graph.

**Chord datasets not included in ChoCo.** Although other collections providing harmonic information exist in the literature, some of them were currently discarded for the reasons explained below. The *Leadsheet* dataset<sup>50</sup> separately annotates chord progressions for each segment (e.g. intro, chorus) but does not provide information on how structures are laid out in the piece. *GuitarSet*<sup>51</sup> only provides 3 unique (and short) chord progressions. *UMA-Piano*<sup>52</sup> only contains audio recordings of chords, played independently. Finally, *POP909*<sup>53</sup> and the *Kostka-Payne* corpus<sup>54</sup> provide computationally-extracted chords and keys, whereas the first release of ChoCo focuses on high-quality annotations for time being.

## 2.2 From chordal data to JAMS datasets

The first challenge of bringing together existing chord datasets into a coherent, uniform corpus is the variety of formats in which chord annotations, and other related information, are encoded. In order to address this issue, we leverage the popular JAMS data structure<sup>13</sup> as a simple, content-agnostic wrapper for expressing music annotations in general, and chord annotations in particular. JAMS relies on the popular Web data exchange JSON format, and enforces the following structure based on three basic properties (see <https://jams.readthedocs.io/> for additional details):

1. *file\_metadata*, describing the music piece these annotations refer to. More precisely, it contains these properties:
  - *identifiers*, optionally providing explicit links to external resources, mostly relating to cataloguing information from e.g. MusicBrainz (<https://musicbrainz.org>);
  - *artist*, *title*, *release*, where the former usually refers to the performer or band, and the latter is intended as a more general definition of album;
  - *duration*, defining a temporal span within which annotations can fall.
2. *annotations*, a container of annotation objects, each describing a specific *namespace* (the term *namespace* in JAMS has a different sense than a Web namespace) that identifies the type of the annotation’s subject (e.g., chords, structural segments, emotions, patterns, keys, etc.). These annotations also include metadata to document the annotation process (e.g. whether the annotation is manually produced or inferred by an algorithmic method, the name of the annotator or software, information about the annotation tools, rules and validation).
3. *sandbox*, described as an unrestricted place to store any additional data.

Listings 1 and 2 show excerpts of an example JAMS file from the Isophonics collection<sup>31</sup> annotating chords for Queen’s *Bohemian Rhapsody*, taken from the Isophonics collection.

Although JAMS has an implicit focus for audio-based annotations, its definition and structure are flexible enough to be easily extendable to the symbolic domain. This is also confirmed by the modular design of the codebase, where additional namespaces can be registered by a user, by simply providing regular expressions to validate the annotation content (e.g. a new chord notation). In other words, any arbitrary music annotation can be described within JAMS as long as the atomic observations (e.g. the individual occurrences of chords making up the progression) are described in terms of: *time*, a temporal anchor specifying the onset of the observation; *duration*, *value* (e.g. *Bb:maj7*), and *confidence*, a scalar in  $[0, 1]$  expressing a level of certainty by the annotator (or algorithm). Therefore, the only elements distinguishing audio from symbolic annotations, are the temporal specifications (time and duration), which are described in absolute (seconds) or metrical (measure and beat/offset) terms, respectively.

```

1 {
2   "sandbox": {},
3   "annotations": [
4     {
5       "data": [
6         {
7           "duration": 0.459,
8           "confidence": 1.0,
9           "value": "N",
10          "time": 0.0
11        },
12        {
13          "duration": 3.663,
14          "confidence": 1.0,
15          "value": "Bb:maj6",
16          "time": 0.459
17        },
18        {
19          "duration": 0.789,
20          "confidence": 1.0,
21          "value": "C:7",
22          "time": 4.122
23        },
24        ...
25      ],

```

**Listing 1.** Excerpt of the three first chords in a JAMS file annotating Queen’s Bohemian Rhapsody.

```

1   "annotation_metadata": {
2     "annotation_tools": "",
3     "curator": {
4       "name": "Matthias Mauch",
5       "email": "m.mauch@qmul.ac.uk"
6     },
7     "annotator": {},
8     "version": 1.0,
9     "corpus": "Isophonics",
10    "annotation_rules": "",
11    "validation": "",
12    "data_source": ""
13  },
14  "namespace": "chord",
15  "sandbox": {}
16 }, ... ],
17 "file_metadata": {
18   "jams_version": "0.2.0",
19   "title": "01 Bohemian Rhapsody",
20   "identifiers": {},
21   "release": "",
22   "duration": 358.293,
23   "artist": "Queen"
24 }
25 }

```

**Listing 2.** Annotation and file metadata in a JAMS file annotating Queen’s Bohemian Rhapsody.

### 2.2.1 JAMification of datasets

Considering the diversity of annotation formats and conventions for data organisation (the way content is scattered across folders, files, database tables, etc.), each chord dataset in ChoCo (c.f. Table 1) undergoes a standardisation process leading to the creation of a JAMS dataset. This is needed to aggregate all relevant annotations of a piece (chord, key, etc.) in a single JAMS file, and to extract content metadata from the relevant sources.

The content metadata of a (music) dataset is indeed crucial to identify, describe and retrieve the actual musical content being annotated. This typically includes the title of each piece, artists (composers and/or performers), and cataloguing information (album/release or collected work), ideally with the provision of identifiers (e.g. MusicBrainz IDs). Nevertheless, only the *Mozart Piano Sonata* collection<sup>6</sup> provides complete content metadata in a csv file, as usually expected from a music dataset. When content metadata is missing, this may be found online (HTML pages, supplementary material), from articles/reports documenting the collection, by resolving any cross-reference among files and dataset-specific identifiers, extracted from the actual score (or better, the dataset-specific representation of the score). Alternatively metadata can be derived from the organisation of files in folders. For example, Michael Jackson/Essential Michael Jackson [Disc 01]/1-16\_Beat\_it.lab indicates author, album, disc, track number and title, respectively. This organisation varies as the datasets vary – a consequence of the lack of a standard “datasheet for datasets” in the music domain<sup>55</sup>.

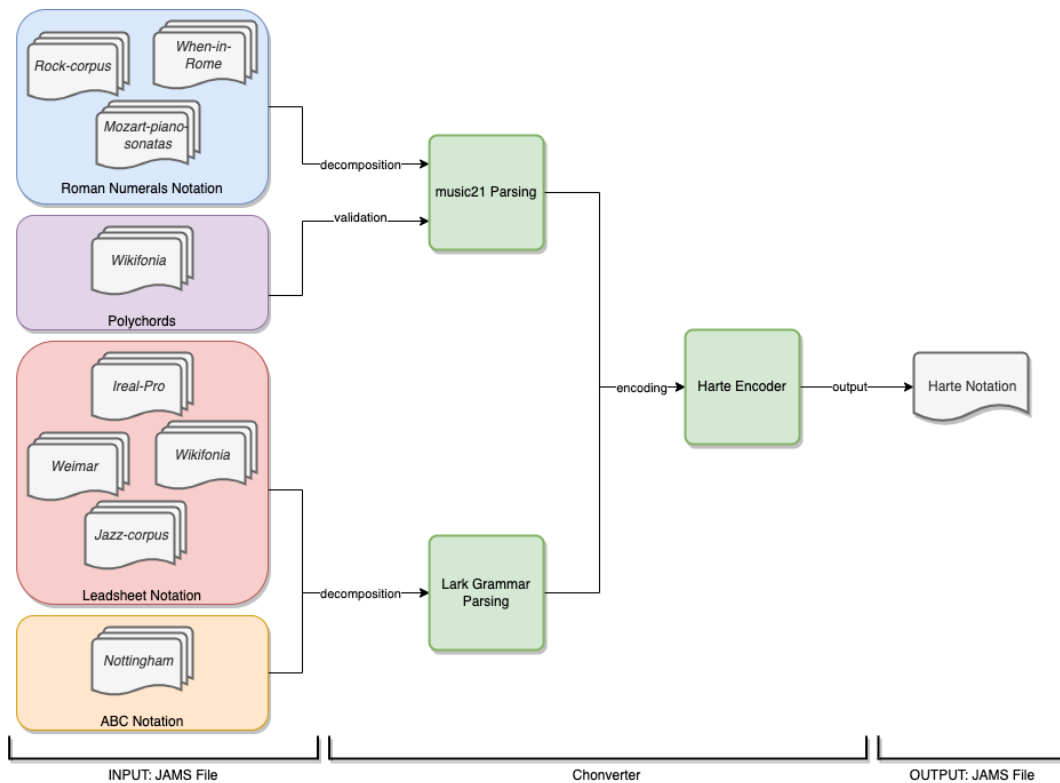
The same issue applies to the extraction, pre-processing, and standardisation of harmonic annotations from these collections, some of which were never released as chord datasets (*Weimar Jazz Database*, *Wikifonia*, *iReal Pro*, *Nottingham*). Harmonic annotations can be encoded in different formats (LAB, XLAB, RomanText, CSV, DCMLab, JSON, SQL, TXT), or extracted from symbolic music (MusicXML, ABC) and backing tracks in proprietary encodings (iReal, MGU). As each collection shows a specific combination of the mentioned issues (different organisation of content and metadata, different annotation formats and conventions), this step required considerable effort. The result of this standardisation process may improve the usability of these resources for music researchers, and simplifies the KG construction process.

Following the standardisation process, each of these 18 JAMS datasets represents a novel contribution per se, due to the heterogeneity of annotation formats and practises, and the limited availability of content metadata in their original version. This also includes *CASD*, a collection that provides chords in JAMS format, but lacks local key annotations, which were retrieved from *Billboard* (we remind that *CASD* is a subset of *Billboard*).

### 2.2.2 Conversion of chord notations

As shown in Table 1, the third element of divergence besides annotation formats and provision of content metadata, is the notation system used to represent chords. To address this issue we perform the following actions: (i) decomposition of domain-specific notations to chord constituting elements; (ii) conversion of the decomposed chord to the Harte framework; (iii) conversion of chord progressions by iteratively applying steps (i) and (ii) to all the chords in a sequence/progression. This yields a new JAMS file with the converted chord annotations.

For all the above steps, specific software was developed for processing the different annotation types contained in the original datasets. There are three main types of chords that are processed: Roman Numerals chords (e.g. C minor: viio7/V),



**Figure 3.** Overview of the Chonverter workflow, describing how different chord notations are converted to the Harte notation.

Polychords (e.g.  $E4, G\#4, B4$ ), Leadsheet chords (e.g.  $Gm7/F$ ). With *Leadsheet chords* we refer to a broader category, although each dataset using this format proposes a different flavour of this notation. For example, a G minor chord in *Wikifonia* is annotated as  $G_{min}$ , whereas the same chord is annotated as  $G-$  in the *Jazz-corpus*.

As outlined in Table 1, each dataset uses a flavour of the same notation to represent chords, with the exception of *Wikifonia*, where some annotations use both Leadsheet and Polychords even for the same progression; and the Jazz Corpus, providing chords encoded in both Roman Numerals and Leadsheet. Figure 3 provides a taxonomy of the different notational flavours, together with a schematic overview of the conversion workflow.

In step (i), a chord is first decomposed into its components (e.g.  $C_{major} \rightarrow C, E, G$ ). For this purpose, the *Chonverter* uses a family of tools depending on the source notation. Roman numerals are decomposed using the *roman module* of *music21*<sup>56</sup>, a Python library for computational musicology. As Polychords already provide note constituents by definition, this step here is limited to preprocessing the symbols associated to the different pitches in a chord. Polychords are usually mixed with chords annotated in other notations (e.g. Leadsheet), so it is necessary to differentiate the type of chords when parsing. Finally, for each Leadsheet flavour, a context free grammar was created to parse the original annotation of the chord. A different grammar was created for each dataset containing annotations in leadsheet format, namely *Weimar Jazz Database*, *Wikifonia*, and *iReal Pro*, using the *Lark* library (<https://github.com/lark-parser/lark>). Notably, the ABC notation used in *Nottingham* is similar to the Leadsheet notation and was therefore processed in the same way. This process is more intuitive for all collections natively using the Harte notation, as the latter provides sufficient expressiveness for the description of the constituting elements of chords<sup>30</sup>.

After all chords are decomposed as lists of pitches, it is then possible to associate a shorthand (a string) to each list according to the Harte notation (Step (ii)). The *Chonverter* achieves this via *music21* and defines rules for composing Harte chords.

New JAMS files are produced after the last step, each providing a new annotation (with `chord_harte` as namespace). Whenever an original annotation uses *Leadsheet* or *Polychord* notations, the new annotation replaces the original, since the conversion provides a generalisation of the former following a syntactic transformation. Instead, if the original annotation contains *Roman Numerals* chords, the new (converted) annotation is added to the existing one, since the Roman Numerals contain information that would otherwise be lost, i.e. the harmonic functions that the chords hold within the piece.

The Choverter module performs a syntactic conversion of chord labels. However, converting Roman Numeral also requires taking into account the key of the song. Moreover, a distinction has to be made between key-relative and absolute chords.



Some music is always played in the same key, while other pieces are frequently transposed. For example, symphonies are often performed in a fixed key, while lieder are typically performed in multiple keys depending on the singer’s vocal range. Datasets like *When in Rome* contain transcriptions of these key-flexible works. Even in these cases, chords in ChoCo are always converted by taking into account the tonality provided by the original dataset for that piece. However, whenever this happens, the generated conversion, although correct, may only be one of several possible conversions.

### 2.3 The JAMS Ontology and the Choco Knowledge Graph

To represent JAMS annotations as linked data (LD) we design an ontology that formally represents the JAMS data model. The JAMS Ontology is part of the Polifonia Ontology Network (PON), from which we reuse other classes and properties. For consistency, we also reuse the Chord Ontology<sup>23</sup> for modelling chords according to the Harte data model. To represent chords expressed in Roman Numerals instead, we developed the Roman Chord Ontology. Table 2 lists the relevant links to retrieve ChoCo’s resources, including the JAMS Ontology and the Knowledge Graph (KG).

The JAMS Ontology formally defines the semantics of music annotations that are encoded using JAMS. To improve compliance with the ontology and facilitate the generation of linked data, we have established conventions for including relevant information in the creation phase of the JAMS files. In essence, the JAMS Ontology ameliorates the limitations of the current JAMS model, mainly on two fronts: (i) at the level of metadata, enabling the alignment and linking of tracks belonging to different datasets, and also, with external resources available on the Web; (ii) at the annotation level, allowing to describe data (e.g. a chord) by semantically annotating its components (e.g. root, quality, inversions, etc.) rather than using a string label.

Concerning the first level, the JAMS Ontology provides support to trace the original source of an annotation, i.e. whether it refers to a signal representation (audio) or to a symbolic representation (score). This also allows to correctly interpret the content of the annotation. In fact, temporal information is expressed in seconds for audio tracks, whereas it is expressed in beats and measures for symbolic music. This distinction also allows to disambiguate the roles of composers and performers – rather than simply referring to them as “artists”. Furthermore, the alignment with other ontologies makes it possible to describe the musical content of JAMS annotations thereby enabling different types of music analyses.

To enable these functionalities, additional data was dumped by the JAMifier in the *Sandbox* of each JAMS file, and new annotation types were created by contributing new *namespaces*. The JAMS Ontology provides a common conceptual, formal model to interpret JAMS annotations and to put their entities in relation with each other. The JAMS Ontology is available online at the following URI:

<https://w3id.org/polifonia/ontology/jams/>

Our ontological requirements can be summarised as follows:

- the resulting KG must represent JAMS files and JAMS annotations as such, including their provenance and process-related information: e.g. source dataset, annotator, confidence, etc;
- the KG must distinguish the performers, who are properties of a track/score, from the composers/authors of a track, which are properties of its corresponding musical composition;
- temporal information must be expressed according to the type of the annotation’s subject, i.e. audio or score;
- chords must be represented according to the data model of these notation families: *Harte* and *Roman Numerals*.

To model this ontology, we reused the *Music Annotation Pattern*<sup>57</sup>, an Ontology Design Pattern (ODP)<sup>58</sup> for modelling different types of music annotations and their related time references. We remark that the terminology used in the JAMS documentation (<https://jams.readthedocs.io/en/stable/>) is adopted to define the JAMS Ontology vocabulary. In particular, the following terms are (re-)used:

- **Annotation:** an annotation is defined as a group of *Observations* (see below) that share certain elements, such as the method used and the type of annotation’s subject (e.g. chords, notes, patterns);
- **Observation:** an observation is defined as the content of an annotation, and includes all the elements that characterise the observation. For example, in the case of an annotation containing chords, each observation corresponds to a chord, and specifies, in addition to the chord value, the temporal information and its confidence.

To develop the JAMS Ontology we apply the eXtreme Design methodology<sup>59</sup> and, according to it, we define a set of competency questions (CQs) that the ontology shall address. These are listed in Table 3. The above CQs were converted into SPARQL queries that served to iteratively test the ontology during its implementation. All SPARQL queries are available in the JAMS Ontology repository (<https://github.com/polifonia-project/jams-ontology>).

| ID   | Competency question                                                                                                |
|------|--------------------------------------------------------------------------------------------------------------------|
| CQ1  | What is the content of the observations contained in a JAMS Annotation?                                            |
| CQ2  | Who is the composer of a musical object?                                                                           |
| CQ3  | Who is the performer of a musical object?                                                                          |
| CQ4  | Who/what is the annotator of an annotation/observation, and what is its type?                                      |
| CQ5  | What is the time frame addressed by an annotation, within a musical object?                                        |
| CQ6  | What is its start time (i.e. the starting time of the time frame)?                                                 |
| CQ7  | Which are the observations included in an annotation?                                                              |
| CQ8  | Given an observation, what is the starting point of the time frame it addresses, within its target musical object? |
| CQ9  | Given an observation, what is its addressed time frame, within its target musical object?                          |
| CQ10 | What is the key of a composition/performance?                                                                      |
| CQ11 | What is the value of an observation?                                                                               |
| CQ12 | What is the confidence of an observation?                                                                          |
| CQ13 | What are the chords of a composition/performance?                                                                  |

**Table 3.** Competency questions (CQs) addressed by the JAMS Ontology.

Figure 4 shows the fragment of the JAMS ontology modelling a JAMS Annotation. Classes and properties from the *Polifonia Ontology Network (PON)* are reused for the definition of artist metadata; in particular, from the PON modules *core*, *musical feature*, *musical composition* and *musical performance*.

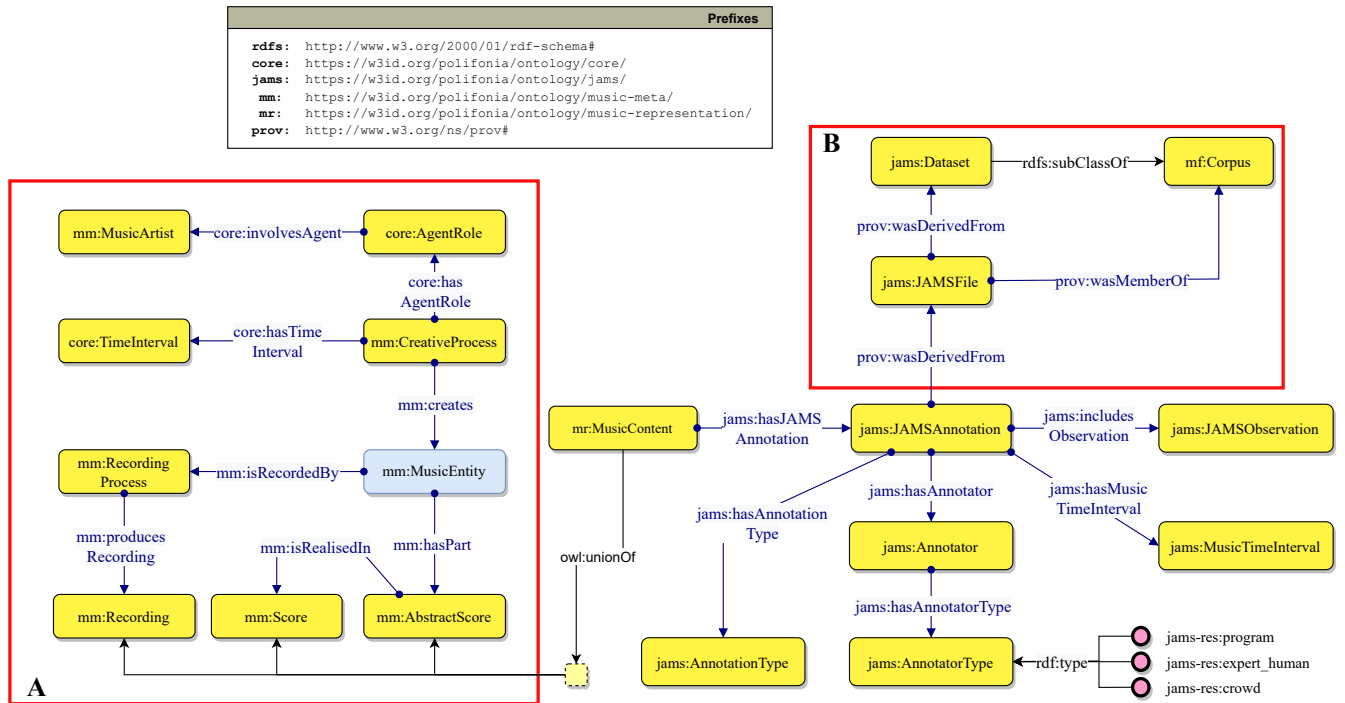
On the left (box A), we define the classes for representing scores (`mf:Score`), recordings (tracks) (`mp:Recording`), and musical compositions (`mc:MusicalComposition`). This is an important distinction between musical objects as the performance of a musical composition can be recorded in a audio file (recording) and/or transcribed in a score. To express these relations, the JAMS ontology provides the properties `mp:recordingOf` and `mc:hasScore`, respectively. Musical compositions are a potential hub for linking data from other musical knowledge graphs (see Section 3.1). The class `core:Agent` is used to relate authors/composers to musical compositions (`mc:hasComposer`) and performers to scores or recordings (`mp:performedBy`).

The Provenance Ontology<sup>60</sup> is reused to model the provenance of JAMS annotations (box B). Each JAMS annotation derives from a JAMS file `jams:JAMSFile` that is either taken or derived (for example, translated from a file in a different format to the JAMS format) from a corpus `mf:Corpus`.

A core class of the JAMS Ontology is `jams:JAMSAnnotation`. It captures the annotation, from a file encoded with the JAMS format, on a musical object (its target): either a recording or a score. A JAMS annotation entity and its musical object are related by means of the property `jams:hasJAMSAnnotation`. An annotation is performed by an annotator `jams:Annotator`, has a time validity `jams:hasMusicTimeInterval`, and contain information of a certain type `jams:AnnotationType` (e.g. chords, keys, etc.). The validity indicates to which time frame, within a musical object, the annotation refers. For example, if an annotation reports the observation of a certain key, that key refers to a segment of the target musical object. Annotators may be of different types (e.g. human annotator, crowd, program), defined by the class `jams:AnnotatorType`. Finally, a `jams:JAMSAnnotation` is composed of a set of observations `jams:JAMSObservation`. Figure 5 depicts the JAMS Ontology fragment that models JAMS observations.

A key aspect of observations and annotations is the identification of the musical object fragment they refer to. We model musical object fragments as musical time intervals `jams:MusicTimeInterval`. Musical time intervals can be expressed in different ways, depending on the type of musical object. For example, if the subject of an observation (and in turn of an annotation) is a recording, then we most probably identify its fragments in terms of seconds. If we deal with scores, we may want to use a combination of measures and beats. To make the ontology as flexible as possible for expressing musical time intervals, we model them as being defined by musical time indexes (`jams:MusicTimeIndex`). Each musical time interval has a start time index and an end time index (plus potentially infinite internal time indexes). A musical time index is defined by one or more components (`jams:MusicTimeIndexComponent`), each substantiated by a value (`jams:hasValue`) and a value type (`jams:MusicTimeValueType`). A musical time interval also has a duration (`jams:MusicTimeDuration`) which is expressed by means of a value and a value type (usually seconds for recordings and beats for scores).

Figure 6 shows an example of data annotated using the *JAMS Ontology*. The data are from *wikifonia\_39*, a file from the *Wikifonia* dataset. To read the graph, we suggest to start from the individual highlighted by a red box: the individual `pon-res:AutumnInRomeComposition`, representing a song entitled "*Autumn in Rome*". The file includes two annotations (`pon-res:Wikifonia39KeyAnnotation` and `pon-res:Wikifonia39ChordAnnotation`), derived from a music score (`pon-res:AutumnInRomeComposition mc:hasScore pon-res:AutumnInRomeScore`), therefore their temporal information is expressed as a combination of beats and measures. The chord annotation



**Figure 4.** Fragment of the JAMS ontology describing JAMS files and their provenance, musical objects and JAMS annotations. We use the Graffoo notation (<https://essepuntato.it/graffoo/>) : yellow boxes are classes, blue/green arrows are object/datatype properties, purple circles are individuals, green polygons are datatypes.

(`pon-res:Wikifonia39ChordAnnotation`) contains two observations, the first starting at the beginning of the first measure (*measure 1, beat 1*), while the second starts at the beginning of the second measure (*measure 2, beat 1*). They both have a duration of 3 *beats*.

We remark that the music time interval of an annotation is different, though dependent on, the time interval of its observations: it must include all of them. In the example of Figure 6, the time interval of `pon-res:Wikifonia39ChordAnnotation` starts from the beginning of the first measure (*measure 1, beat 1*) and has a duration of 108 *beats*.

A JAMS observation, according to the JAMS data model, can only have one subject (`jams:hasSubject`), which is a musical feature (`mf:MusicalFeature`) e.g. chord, key mode, pitch. The main musical feature currently treated in ChoCo is the chord. A chord (`mf:Chord`) is indeed modelled as a special type of `mf:MusicalFeature`. We define the property `jams:hasChordSubject` as a subproperty of `jams:hasSubject` to relate JAMS chord observations to their subjects.

ChoCo and the JAMS Ontology address the most commonly used chord notations: Harte and Roman Numerals. For modelling chords encoded according to the Harte notation we reuse the Chord ontology<sup>23</sup>. We develop an ontology for modelling chords expressed according to the Roman Numerals convention, the *Roman Chord Ontology* (<https://github.com/polifonia-project/roman-chord-ontology>). Figure 7 shows the main classes and properties of the Roman Chord Ontology, which is available at the following URI:

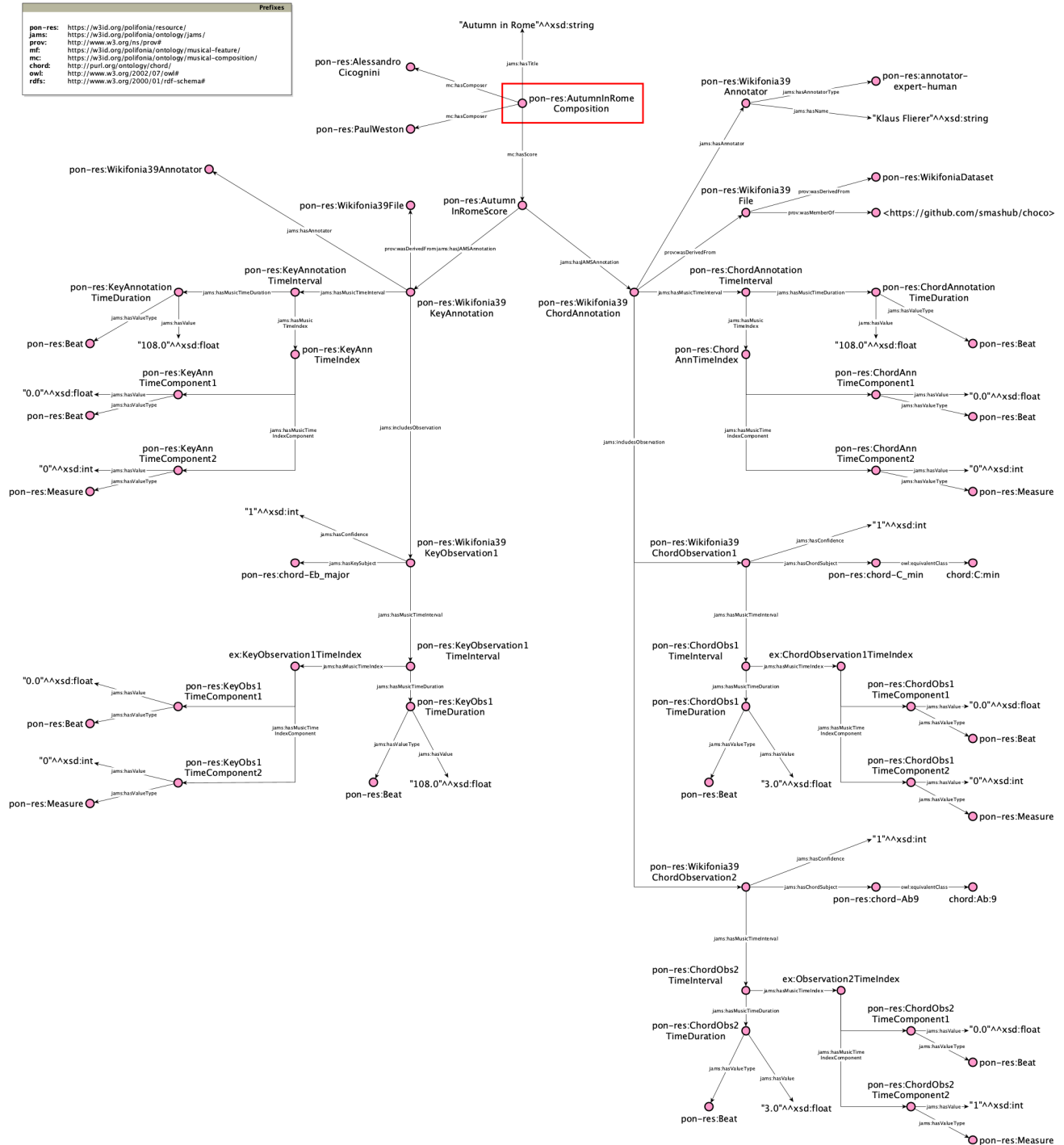
<https://w3id.org/polifonia/ontology/roman-chord/>

The core class `roman:Chord` defines roman numeral chords. A chord is a complex structure, therefore it is described by means of several properties. The classes `roman:BasicFunction` and `roman:Quality` describe the chord from a functional harmony perspective and the quality of the chord (e.g. major, minor, augmented), respectively. The class `roman:Note` describes the absolute pitch of the bass note, while the class `roman:Interval` is used to describe the bass, the internal intervals of the chord and any missing intervals. Each interval is described by the datatype properties `roman:hasDegree`, which describes the degree of the interval, and `roman:hasModifier`, which describes any alterations to the interval. Finally, the datatype property `roman:inversionType` defines the possible type of inversion of the chord.

We release service APIs to generate knowledge graphs of roman numeral chords, starting from its symbol and a reference key. The API service can be queried as follows:

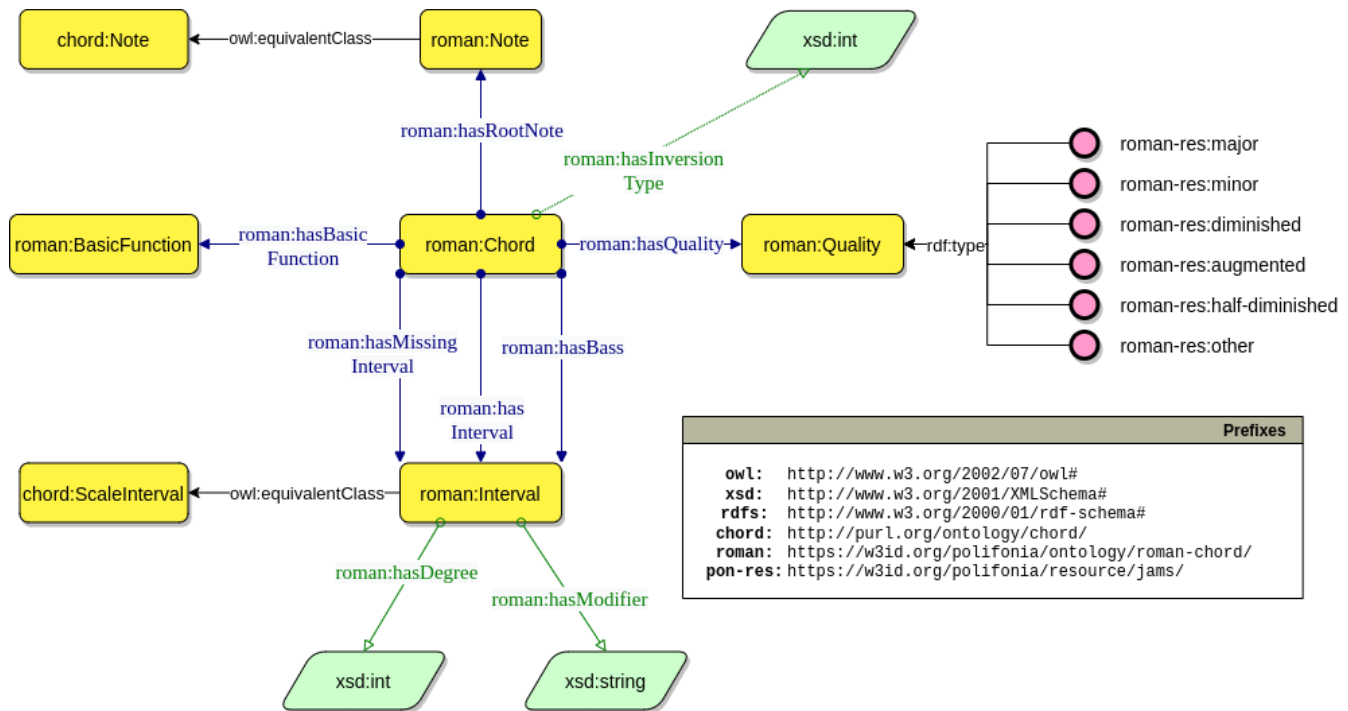
[https://w3id.org/polifonia/resource/roman-chord/\[romanChord\]\\_\[key\]](https://w3id.org/polifonia/resource/roman-chord/[romanChord]_[key])





**Figure 6.** Example of data modelled using *JAMS Ontology*, extracted from a track from the *Wikifonia* dataset. The track is annotated from a score, therefore annotations and observations contain time references expressed in *beat* and *measure*.





**Figure 7.** The Roman Chord Ontology describing Roman Numeral Chords and their constituting elements.

duration of pieces, which also confirms the diversity of musical genres in ChoCo (Table 1). For instance, a folk tune can span a few measures and still possess a musical identity with respect to the genre; in contrast, a sonata can cover hundreds of measures.

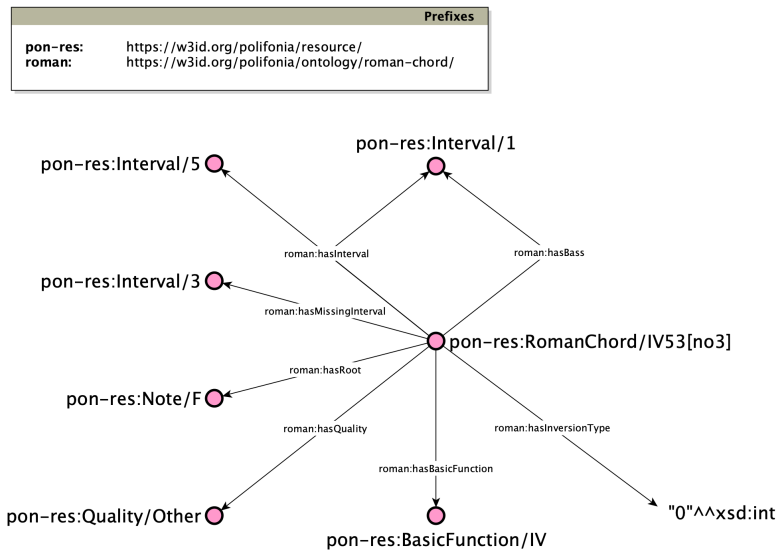
From the metadata extraction of the JAMification step (c.f. Section 2.2), it was possible to disambiguate 2421 artists as *performers* – which represent 12.05% of the dataset, and a total of 7,304 as *composers* (36.36% of ChoCo). This implies that the remaining 51.59% of JAMS files only provide generic *artist* information (with no distinction between composers and performers), whereas another small portion of the dataset – corresponding to the JazzCorpus (0.37% of ChoCo), does not provide any metadata. An overview of the ten most common performers and composers is reported in Figure 10, with “The Beatles” and “Franz Schubert” being the most recurring names, respectively.

The JAMS files in ChoCo also contain 771 links to other resources, representing about 3.8% of the dataset. These were extracted from the original collections, and automatically verified and corrected for validity (link/identifier working) and consistency (disambiguation of the resource pointed, e.g., musical work, recording, and release). Most links point to MusicBrainz (78%), whereas a few of them link to Wikidata (6%), IMSLP (6%), YouTube (5%), and to other datasets (5%).

In addition to these explicit links, which can already be found in the JAMS files, we also link the resources in the ChoCo KG to two other large-scale music datasets on the Web:

1. **MIDI Linked Data Cloud**<sup>28</sup>. The ChoCo chord annotations can be useful for harmonic analyses of existing scores and symbolic music representations, e.g. MIDI. To link MIDI URIs with ChoCo URIs, we compare the string similarity of the original MIDI filename and the JAMS `file_metadata` name, both typically containing the band/artist and song names, and link them through `midi:midiOf` if their similarity is  $>.80$ . This yields 2,411 links. However, we do not inspect the musical content to establish this linkage, meaning that the harmonic annotation of a sonata in C minor would be linked to the same sonata in D minor, as long as their titles are highly similar. Therefore, the verification and the provision of links that are musically plausible (beyond the metadata) are currently under investigation.
2. **Listening Experience Database (LED)**<sup>62</sup>. Relating harmonic properties of pieces and their evolution to music listening experiences throughout history is also another promising direction. For those listening experiences that are explicitly associated to a musical work through `dc:subject` and `mo:performance_of` (where `dc` and `mo` prefix *Dublin Core* and *Music Ontology*, respectively), we extract links with ChoCo’s resources via text similarity of work titles (using the same criteria as before). Links can be further filtered whenever a musical work in LED also provides a reference to the artist (via `mo:composer` or `mo:performer`). Overall, this yields 1996 links.

These additional links open up new research directions, as they allow to relate harmonic content (chord changes, harmonic



**Figure 8.** An example of a Knowledge Graph generated using the *Roman Chord Ontology*, showing the composition of a IV53 [no3] chord.

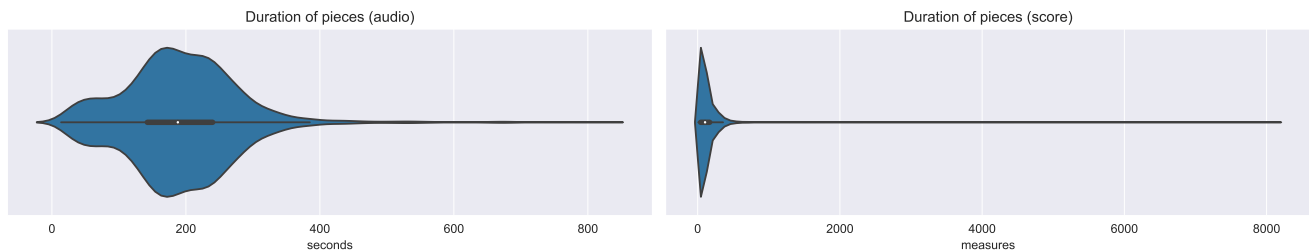
complexity, tension, etc.) to other musical properties that are inherently present in the music (melodic contour, expressive variations, instrumental changes, etc.), or that may have been elicited certain emotions, memories, and feelings in listeners. Here we report an example of a listening experience of “*So What*” in LED (<https://data.open.ac.uk/page/led/lexp/1431335026178>), which was linked to 8 chord annotations in ChoCo.

«What do you mean by playing “without harmony”? Using a pedal tone, which Coltrane got into after a period of very dense harmonic playing. He would use one or two harmonic references throughout a song, as he did on “*So What*” [from Miles Davis’s *Kind of Blue*, on Columbia]. It was basically D for sixteen bars, E flat for eight bars, and then back to D. Ultimately, he worked with only one harmonic reference point, and then in “*Ascension*” [from *Best of John Coltrane: His Greatest Years*, on Impulse] there was nothing harmonically.» (Steve Kuhn in “*The Great Jazz Pianists: Speaking of Their Lives and Music*”)

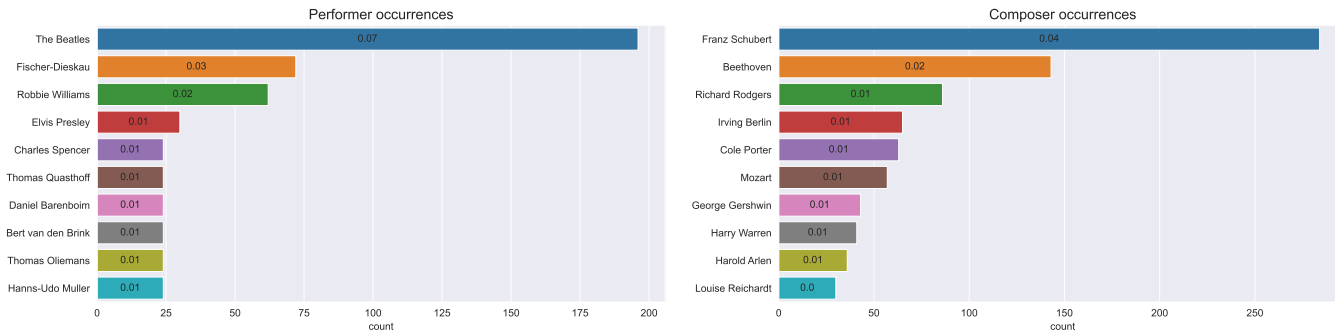
### 3.2 Overview of chordal annotations

This section provides statistics on the content of chord annotations in ChoCo, their observations and temporal onsets; similar statistics can also be extracted for tonality annotations (local and global keys), but are excluded here to focus on chordal content.

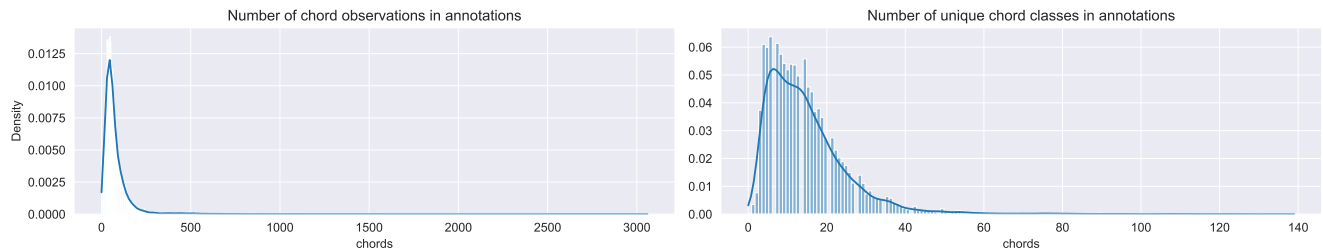
Overall, and without any simplification/collapsing of chords, there are 1,575,409 chord occurrences/observations in ChoCo, with an average annotation having  $76.74 \pm 86.76$  chords (Figure 11, left). When looking at the unique chord occurrences in the harmonic progressions (chord classes) – measuring the chordal diversity of the annotations, the dataset counts 306,407 chords, which are drawn from a set of 7,281 possible classes. An annotation, on average, uses  $14.92 \pm 11.10$  chord classes (Figure 11,



**Figure 9.** Distribution of track (left) and score (right) durations, measured in seconds and measures, respectively.



**Figure 10.** Overview of the ten most common performers (*left*) and composers (*right*) in the dataset, when explicitly distinguishable from their generic “artist” attribution in the metadata.



**Figure 11.** Distribution of the number of chord observations per annotation (*left*) and their distinct chord classes (*right*). The latter can also be considered as the cardinality of the chord set used by each annotation.

*right*). The average duration of chord observations in audio and score JAMS is  $2.85 \pm 12.55$  seconds and  $3.69 \pm 8.25$  beats, respectively. For most statistics reported in this section, we observe right skewed distributions for these variables (long tails on the right side) as negative values (e.g. negative durations) do not occur.

The fifteen most common chords in ChoCo, based on their absolute and relative occurrences, are reported in Figure 12 (*left*). Absolute counts are obtained by accumulating the chord counts for each annotation/progression across the dataset (as if all annotations refer to the same piece). Instead, relative counts are computed by first normalising the absolute counts of each annotation by the number of chord observations in the progression; then averaging the resulting chord frequencies across all annotations. Analogously, Figure 12 (*right*) reports the same statistics after removal of consecutively repeated chords. This pre-processing step aims to mitigate consecutive repetitions (which may arise due to the different temporal granularity of chord observations, or possess a harmonic function) from inflating the chord count. Regardless of the counting method, the three most common chords in the dataset are: *C:maj*, *G:maj*, and *F:maj*.

A similar analysis is also reported for chord n-grams, which are typically used to find harmonic patterns in songs. To avoid trivial n-grams, these are computed after removal of consecutive repetitions (e.g. *G:7*, *G:7*, *C:maj* becoming *G:7*, *C:maj*). Table 4 ranks the first 10 n-grams based on their relative count (frequency).

To conclude, the number of chord annotations for which the identity of the annotators is known is 796 (3.9% of the dataset).

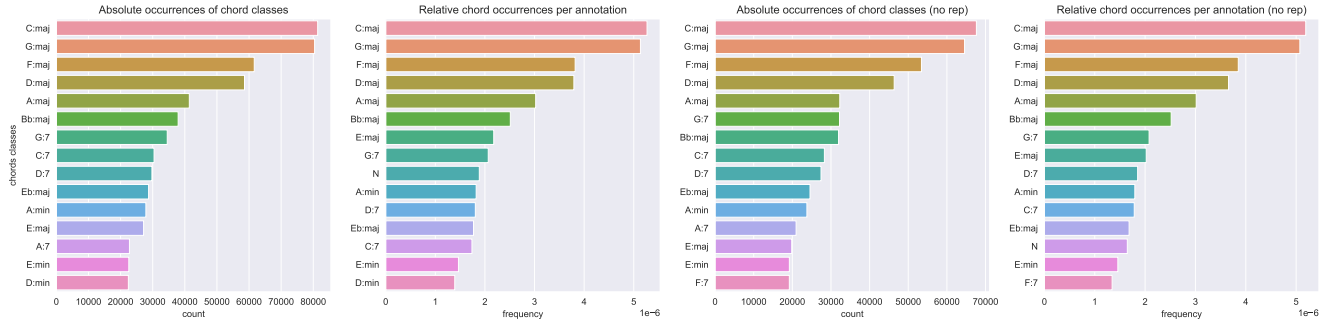
## 4 Technical Validation

To validate the data transformation workflow presented in Section 2 (Figure 1), focusing on the output of the JAMifier (generation of JAMS files from arbitrary chord collections) and the Chonverter (chord alignment and conversion) modules, we conducted two separate analyses: a groundtruth evaluation of JAMS files, and an expert validation of chord conversions.

### 4.1 Validation of the JAMifier

As the goal of the JAMifier is to automatically generate a JAMS dataset given a music collection providing chord annotations and metadata in different formats, notations, and conventions, this first evaluation addresses the following question.

**Q 1** How complete and accurate are ChoCo’s JAMS files – for metadata and harmonic annotations, after the JAMification?



**Figure 12.** Absolute and relative occurrences of chord classes in the original annotations (*left, centre-left*), and after removal of consecutively repeated chords (*right, centre-right*). Absolute occurrences are counted and accumulated throughout the corpus, whereas relative occurrences are first aggregated per annotation, as frequencies, then averaged across the dataset.

To answer this question, we carried out a series of tests to compare a sample of generated JAMS files with those that are expected from this process. This required the creation of a groundtruth dataset of JAMS files that were manually produced by two human annotators from a given template (the backbone of a JAMS file), and through manual inspection of the original collections. For example, given a sample of the Wikifonia subset, the validator was expected to fill the JAMS template by: opening the MusicXML file of each assigned piece; inserting the relevant metadata (title, composer, duration, etc.) into the appropriate fields; and finding the (Leadsheet) chord labels annotated on the score – to create a JAMS *Observation* out of each of them. Annotators were first instructed on the task, and a preliminary annotation trial was performed to assess their reliability. After the trial, annotators received 4 templates for each subset and produced 72 gold JAMS files in total. The corresponding JAMification output is then compared to the groundtruth to measure: (i) the coverage and the accuracy of the metadata; and (ii) the coverage and error of chord and key annotations.

For the metadata, coverage is computed as the proportion of metadata fields in the gold JAMS that can also be found in the generated JAMS, regardless of their values. For example, if *title*, *composers*, *genre*, and *duration* are the expected metadata fields for a given JAMS file, and the generated counterpart only provides records for *title* and *duration*, coverage would account for 0.5 (even if both *title* and *duration* are incorrect). To provide a complementary view, metadata accuracy of common fields is computed as the normalised Levenshtein similarity among the generated and expected values for strings; or as the relative variance from the expected value for numerical fields (e.g. duration). The accuracies are then averaged for each JAMS file.

The results of this evaluation are reported in Table 5, aggregated for each subset and separated from the identifiers that were extracted from the JAMification (e.g. MusicBrainz, Wikidata – c.f. Section 3.1). Overall, maximum accuracy and coverage are attained for most collections, and all the possible identifiers are always extracted with no errors.

For the harmonic annotations in the JAMS files, comparison with the gold counterparts is focused on *coverage* and *error* – reported independently for times (e.g. the onset of a chord occurrence), durations (e.g. how long a chord occurrence spans), and labels (e.g. a *C:maj* chord) of the observations in each annotation. The evaluation is thus in line with the structure of an observation in JAMS’ annotations (see Section 2.2 and Listings 1, 2). In this case, *coverage* measures the amount of the overlap between the generated and the expected observation fields, without taking order into account (this is because an extra observation may have been inserted by the annotator, thus breaking the desired alignment for comparison). For example, if (C:maj, G:maj, D:7, F:maj) and (N, C:maj, G:maj, D:7) are the labels of a generated chord annotation and the corresponding gold, respectively, the silent chord “N” breaks the alignment of those sequences. In this case, coverage would still be 3/4, as all the other chord labels are included in generated annotation. Instead, errors are computed from a 1-to-1 comparison of fields – which are assumed to be aligned. The latter can be reported according to the unit of measure of each field: seconds and beats for *time* and *duration*, and normalised Levenshtein distance for *labels* (string values).

Table 6 reports the results of this last evaluation for both key and chord annotations, where each metric is averaged by subset (mean and standard deviation). Results show good coverage and minimum error for most subsets, thus confirming the quality of the JAMification output. An exception is the Mozart Piano Sonata collection, for which low coverage and high errors are reported for key annotations. After having manually compared the JAMS sample for this subset, we found that the observations annotated by our validators in the gold set used a different temporal granularity (e.g. merging two consecutively repeated observations and aggregating their time and duration), compared to the JAMification output. Although this affected the evaluation results, both these annotations can be deemed equivalent.

| Order | Rank | Chord 1 | Chord 2 | Chord 3 | Chord 4 | Frequency    | Occurrences |
|-------|------|---------|---------|---------|---------|--------------|-------------|
| 2     | 1    | G:maj   | C:maj   | -       | -       | 9.894371e-07 | 11560       |
|       | 2    | C:maj   | G:maj   | -       | -       | 9.314316e-07 | 9968        |
|       | 3    | C:maj   | F:maj   | -       | -       | 8.578674e-07 | 9837        |
|       | 4    | D:maj   | G:maj   | -       | -       | 8.447899e-07 | 11229       |
|       | 5    | G:7     | C:maj   | -       | -       | 8.270923e-07 | 12590       |
|       | 6    | G:maj   | D:maj   | -       | -       | 8.236944e-07 | 9591        |
|       | 7    | F:maj   | C:maj   | -       | -       | 7.588854e-07 | 8547        |
|       | 8    | D:7     | G:maj   | -       | -       | 7.092709e-07 | 10673       |
|       | 9    | A:maj   | D:maj   | -       | -       | 6.319998e-07 | 6925        |
|       | 10   | C:7     | F:maj   | -       | -       | 6.247398e-07 | 10362       |
| 3     | 1    | G:maj   | C:maj   | G:maj   | -       | 4.156081e-07 | 4487        |
|       | 2    | C:maj   | F:maj   | C:maj   | -       | 4.022300e-07 | 4167        |
|       | 3    | D:maj   | G:maj   | D:maj   | -       | 3.518498e-07 | 4473        |
|       | 4    | C:maj   | G:maj   | C:maj   | -       | 3.210295e-07 | 3209        |
|       | 5    | G:maj   | D:7     | G:maj   | -       | 2.757892e-07 | 3411        |
|       | 6    | G:maj   | D:maj   | G:maj   | -       | 2.755515e-07 | 3483        |
|       | 7    | C:maj   | G:7     | C:maj   | -       | 2.685492e-07 | 3371        |
|       | 8    | F:maj   | C:maj   | F:maj   | -       | 2.601499e-07 | 2660        |
|       | 9    | A:maj   | E:maj   | A:maj   | -       | 2.201239e-07 | 1767        |
|       | 10   | A:maj   | D:maj   | A:maj   | -       | 2.151695e-07 | 2450        |
| 4     | 1    | G:maj   | C:maj   | G:maj   | C:maj   | 1.984606e-07 | 1933        |
|       | 2    | C:maj   | G:maj   | C:maj   | G:maj   | 1.897574e-07 | 1746        |
|       | 3    | C:maj   | F:maj   | C:maj   | F:maj   | 1.840459e-07 | 1693        |
|       | 4    | F:maj   | C:maj   | F:maj   | C:maj   | 1.759950e-07 | 1509        |
|       | 5    | D:maj   | G:maj   | D:maj   | G:maj   | 1.647309e-07 | 2256        |
|       | 6    | G:maj   | D:maj   | G:maj   | D:maj   | 1.609514e-07 | 2105        |
|       | 7    | D:7     | G:maj   | D:7     | G:maj   | 1.587393e-07 | 1873        |
|       | 8    | A:maj   | E:maj   | A:maj   | E:maj   | 1.497483e-07 | 998         |
|       | 9    | E:maj   | A:maj   | E:maj   | A:maj   | 1.453102e-07 | 1067        |
|       | 10   | G:7     | C:maj   | G:7     | C:maj   | 1.338413e-07 | 1593        |

**Table 4.** Summary of the most common chord n-grams ( $n = 2, 3, 4$ ), ranked by their relative occurrence (frequency) per chord annotation. The last column reports the corresponding total number of n-gram occurrences in the dataset (no aggregation).

## 4.2 Validation of the Chonverter

Following the data transformation workflow illustrated in Figure 1, we recall that the output of the JAMification step that does not natively provide Harte chord labels undergoes an alignment/conversion process through the *Chonverter*. First, the *Chonverter* aligns chord labels to one of the three chord families introduced in Section 2.2.2, namely: *Leadsheet* (Harte), *Roman*, and *Polychord*. Then, a syntactic conversion is performed on each chord class, independently, to infer the corresponding Harte label. Evaluating the output of the *Chonverter* can thus be formulated as follows.

**Q 2** *How accurate and musically plausible are the chord alignment and chord conversion steps?*

Conversely to the previous evaluation, addressing this question requires musical expertise and familiarity with different chord notations. Therefore, we performed a 2-step evaluation with music experts to validate the alignment and the conversion rules. Four participants with at least 5 years of musical training were recruited for this experiment. Participants were first introduced to the task, and asked to express their level of familiarity with the different chord notations, and the validation methodology. Given the nature of the validation, no personal record was recorded from participants and minimal risk clearance was granted from the Research Ethics Office of King’s College London (registration number: MRSP-21/22-32842).

**Step 1** The first step focused on validating the context-free grammars used to parse chords in the original formats and aligning them to the corresponding chord families. Participants were presented with 3 different grammars, including 250 mapping rules to validate. Whenever a rule was deemed incorrect, participants were asked to provide the expected mapping.



| partition            | metadata            |                     | identifiers         |                     |
|----------------------|---------------------|---------------------|---------------------|---------------------|
|                      | coverage $\uparrow$ | accuracy $\uparrow$ | coverage $\uparrow$ | accuracy $\uparrow$ |
| biab-internet-corpus | $0.95 \pm 0.1$      | $0.9243 \pm 0.0835$ | -                   | -                   |
| billboard            | $1.0 \pm 0.0$       | $1.0 \pm 0.0$       | -                   | -                   |
| chordify             | $1.0 \pm 0.0$       | $1.0 \pm 0.0$       | $1.0 \pm 0.0$       | $1.0 \pm 0.0$       |
| ireal-pro            | $1.0 \pm 0.0$       | $1.0 \pm 0.0$       | -                   | -                   |
| isophonics           | $1.0 \pm 0.0$       | $1.0 \pm 0.0$       | -                   | -                   |
| jaah                 | $0.8036 \pm 0.0595$ | $1.0 \pm 0.0$       | $1.0 \pm 0.0$       | $1.0 \pm 0.0$       |
| jazz-corpus          | $1.0 \pm 0.0$       | $1.0 \pm 0.0$       | -                   | -                   |
| mozart-piano-sonatas | $0.875 \pm 0.0$     | $1.0 \pm 0.0$       | $1.0 \pm 0.0$       | $1.0 \pm 0.0$       |
| nottingham           | $1.0 \pm 0.0$       | $1.0 \pm 0.0$       | -                   | -                   |
| real-book            | $1.0 \pm 0.0$       | $1.0 \pm 0.0$       | -                   | -                   |
| robbie-williams      | $1.0 \pm 0.0$       | $1.0 \pm 0.0$       | -                   | -                   |
| rock-corpus          | $1.0 \pm 0.0$       | $1.0 \pm 0.0$       | -                   | -                   |
| rwg-pop              | $1.0 \pm 0.0$       | $0.9999 \pm 0.0001$ | -                   | -                   |
| schubert-winterreise | $1.0 \pm 0.0$       | $1.0 \pm 0.0$       | $1.0 \pm 0.0$       | $1.0 \pm 0.0$       |
| uspop2002            | $1.0 \pm 0.0$       | $0.9661 \pm 0.062$  | -                   | -                   |
| weimar               | $1.0 \pm 0.0$       | $0.9878 \pm 0.0243$ | $1.0 \pm 0.0$       | $1.0 \pm 0.0$       |
| when-in-rome         | $0.7976 \pm 0.0558$ | $0.9608 \pm 0.0694$ | -                   | -                   |
| wikifonia            | $0.95 \pm 0.1$      | $0.95 \pm 0.1$      | -                   | -                   |

**Table 5.** Average coverage and accuracy of metadata and identifiers in the generated JAMS files, per ChoCo partition. The dash symbol denotes a partition that does not provide any identifiers.

**Step 2** Once chords were converted, the final result of the conversion was validated. This step also allowed for the validation of other conversion types that were not validated in Step 1, such as Roman numerals and Polychords. In addition, even for annotations originally provided in Leadsheet, this step allows for the validation of added/removed notes and inversions.

The former step allowed to validate all the grammar rules used for decomposing leadsheet chords into their constituting degrees. Each grammar consists of a set of *shorthands* grouped into classes. For example, the class referring to minor chords is composed of the shorthands "m" and "min". Each class is then mapped to the degrees that compose that type of chord: for minor chords, the degrees associated with that class are 1, b3, 5. This type of validation was required due to the limited musical background of the dataset’s curators. All grammar rules reported incorrect by the experts were corrected and revised. A total of 27 rules within the validated grammars were updated. The corrections were of two main types: i) *correct shorthands but incorrect degrees*: the group of shorthands assigned to degrees was correct, but the degrees into which the chord was decomposed had one or more errors; ii) *inconsistent group of shorthands*: the grouping of shorthands in classes was incorrect. In this case, the shorthand(s) not belonging to the class was moved to the correct class if it existed, otherwise a new class was created. This implies that the preliminary chord alignment of the *Chonverter* is potentially error free.

The second validation step consisted in distributing spreadsheets in which the original chords were shown in the first column whereas the second column showed the chords converted by the *Chonverter* module. Before starting this validation phase, all participants were provided with a thorough documentation of all types of annotation used, including Harte. Furthermore, chords annotated in the *Roman Numeral* format, which had not been validated in the previous step, were tested for the first time. Experts were asked to mark whether the conversion to the Harte format was correct or not. The evaluation results are as the percentage of corrected chords out of the total (Table 7).

## 5 Usage Notes

The availability of a large chord dataset, providing high-quality harmonic annotations with temporal information, content metadata, and links to external resources, is of considerable interest to several research communities. In the field of MIR, chord datasets are a fundamental prerequisite for training and evaluating content-based music algorithms that can accommodate a variety of tasks – from chord recognition and cover song detection, to automatic composition systems. For musicology and computational music analysis, the scale and diversity of ChoCo<sup>29</sup> would enable large scale cross-corpus studies across different musical periods, genres, and artists (e.g. uncovering potential influences), and the KG can also be leveraged to run complex queries entailing certain musicological properties of chords, rather than relying exclusively on their notation-specific label.

| partition            | type  | Key coverages $\uparrow$ |               |                | Key errors $\downarrow$ |                    |                 | Chord coverages $\uparrow$ |                 |                 | Chord errors $\downarrow$ |                 |                 |
|----------------------|-------|--------------------------|---------------|----------------|-------------------------|--------------------|-----------------|----------------------------|-----------------|-----------------|---------------------------|-----------------|-----------------|
|                      |       | time                     | duration      | label          | time                    | duration           | label           | time                       | duration        | label           | time                      | duration        | label           |
| billboard            | audio | 1.0 $\pm$ 0.0            | 1.0 $\pm$ 0.0 | 1.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0           | 0.0 $\pm$ 0.0      | 0.0 $\pm$ 0.0   | 1.0 $\pm$ 0.0              | 1.0 $\pm$ 0.0   | 1.0 $\pm$ 0.0   | 0.0 $\pm$ 0.0             | 0.0 $\pm$ 0.0   | 0.0 $\pm$ 0.0   |
| chordify             | audio | 1.0 $\pm$ 0.0            | 1.0 $\pm$ 0.0 | 1.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0           | 0.0 $\pm$ 0.0      | 0.0 $\pm$ 0.0   | 1.0 $\pm$ 0.0              | 1.0 $\pm$ 0.0   | 1.0 $\pm$ 0.0   | 0.0 $\pm$ 0.0             | 0.0 $\pm$ 0.0   | 0.0 $\pm$ 0.0   |
| isophonics           | audio | 1.0 $\pm$ 0.0            | 1.0 $\pm$ 0.0 | 1.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0           | 0.0 $\pm$ 0.0      | 0.0 $\pm$ 0.0   | 1.0 $\pm$ 0.0              | 1.0 $\pm$ 0.0   | 1.0 $\pm$ 0.0   | 0.0 $\pm$ 0.0             | 0.0 $\pm$ 0.0   | 0.0 $\pm$ 0.0   |
| jaah                 | audio | 1.0 $\pm$ 0.0            | 1.0 $\pm$ 0.0 | 1.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0           | 0.0 $\pm$ 0.0      | 0.0 $\pm$ 0.0   | 0.95 $\pm$ 0.1             | 0.95 $\pm$ 0.1  | 1.0 $\pm$ 0.0   | 0.06 $\pm$ 0.13           | 0.06 $\pm$ 0.13 | 0.0 $\pm$ 0.0   |
| robbie-williams      | audio | 1.0 $\pm$ 0.0            | 1.0 $\pm$ 0.0 | 1.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0           | 0.0 $\pm$ 0.0      | 0.0 $\pm$ 0.0   | 1.0 $\pm$ 0.0              | 1.0 $\pm$ 0.0   | 1.0 $\pm$ 0.0   | 0.0 $\pm$ 0.0             | 0.0 $\pm$ 0.0   | 0.0 $\pm$ 0.0   |
| rwc-pop              | audio | -                        | -             | -              | -                       | -                  | -               | 1.0 $\pm$ 0.0              | 0.53 $\pm$ 0.45 | 1.0 $\pm$ 0.0   | 0.0 $\pm$ 0.0             | 0.0 $\pm$ 0.0   | 0.0 $\pm$ 0.0   |
| schubert-winterreise | audio | 1.0 $\pm$ 0.0            | 1.0 $\pm$ 0.0 | 1.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0           | 0.0 $\pm$ 0.0      | 0.0 $\pm$ 0.0   | 1.0 $\pm$ 0.0              | 1.0 $\pm$ 0.0   | 1.0 $\pm$ 0.0   | 0.0 $\pm$ 0.0             | 0.0 $\pm$ 0.0   | 0.0 $\pm$ 0.0   |
| uspop2002            | audio | -                        | -             | -              | -                       | -                  | -               | 1.0 $\pm$ 0.0              | 0.3 $\pm$ 0.26  | 1.0 $\pm$ 0.0   | 0.0 $\pm$ 0.0             | 0.0 $\pm$ 0.0   | 0.0 $\pm$ 0.0   |
| weimar               | audio | 1.0 $\pm$ 0.0            | 1.0 $\pm$ 0.0 | 1.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0           | 0.0 $\pm$ 0.0      | 0.0 $\pm$ 0.0   | 1.0 $\pm$ 0.0              | 1.0 $\pm$ 0.0   | 1.0 $\pm$ 0.0   | 0.0 $\pm$ 0.0             | 0.0 $\pm$ 0.0   | 0.0 $\pm$ 0.0   |
| biab-internet-corpus | score | 1.0 $\pm$ 0.0            | 1.0 $\pm$ 0.0 | 1.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0           | 0.0 $\pm$ 0.0      | 0.0 $\pm$ 0.0   | 0.95 $\pm$ 0.1             | 1.0 $\pm$ 0.0   | 1.0 $\pm$ 0.0   | 0.05 $\pm$ 0.1            | 0.0 $\pm$ 0.0   | 0.0 $\pm$ 0.0   |
| ireal-pro            | score | 1.0 $\pm$ 0.0            | 1.0 $\pm$ 0.0 | 1.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0           | 0.0 $\pm$ 0.0      | 0.0 $\pm$ 0.0   | 1.0 $\pm$ 0.0              | 1.0 $\pm$ 0.0   | 1.0 $\pm$ 0.0   | 0.0 $\pm$ 0.0             | 0.0 $\pm$ 0.0   | 0.0 $\pm$ 0.0   |
| jazz-corpus          | score | 1.0 $\pm$ 0.0            | 1.0 $\pm$ 0.0 | 1.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0           | 0.0 $\pm$ 0.0      | 0.0 $\pm$ 0.0   | 1.0 $\pm$ 0.0              | 1.0 $\pm$ 0.0   | 1.0 $\pm$ 0.0   | 0.0 $\pm$ 0.0             | 0.0 $\pm$ 0.0   | 0.0 $\pm$ 0.0   |
| mozart-piano-sonatas | score | 0.5 $\pm$ 0.58           | 0.0 $\pm$ 0.0 | 0.5 $\pm$ 0.58 | 62.55 $\pm$ 125.03      | 139.75 $\pm$ 83.75 | 0.25 $\pm$ 0.29 | 0.85 $\pm$ 0.3             | 0.88 $\pm$ 0.25 | 0.75 $\pm$ 0.5  | 0.25 $\pm$ 0.5            | 0.15 $\pm$ 0.19 | 0.15 $\pm$ 0.3  |
| nottingham           | score | 1.0 $\pm$ 0.0            | 1.0 $\pm$ 0.0 | 1.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0           | 0.0 $\pm$ 0.0      | 0.0 $\pm$ 0.0   | 0.75 $\pm$ 0.25            | 1.0 $\pm$ 0.0   | 1.0 $\pm$ 0.0   | 0.85 $\pm$ 0.6            | 0.0 $\pm$ 0.0   | 0.0 $\pm$ 0.0   |
| real-book            | score | 1.0 $\pm$ 0.0            | 1.0 $\pm$ 0.0 | 1.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0           | 0.0 $\pm$ 0.0      | 0.0 $\pm$ 0.0   | 1.0 $\pm$ 0.0              | 1.0 $\pm$ 0.0   | 1.0 $\pm$ 0.0   | 0.0 $\pm$ 0.0             | 0.0 $\pm$ 0.0   | 0.0 $\pm$ 0.0   |
| rock-corpus          | score | 1.0 $\pm$ 0.0            | 1.0 $\pm$ 0.0 | 1.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0           | 0.0 $\pm$ 0.0      | 0.0 $\pm$ 0.0   | 1.0 $\pm$ 0.0              | 1.0 $\pm$ 0.0   | 1.0 $\pm$ 0.0   | 0.0 $\pm$ 0.0             | 0.0 $\pm$ 0.0   | 0.0 $\pm$ 0.0   |
| schubert-winterreise | score | 1.0 $\pm$ 0.0            | 1.0 $\pm$ 0.0 | 1.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0           | 0.0 $\pm$ 0.0      | 0.0 $\pm$ 0.0   | 1.0 $\pm$ 0.0              | 1.0 $\pm$ 0.0   | 1.0 $\pm$ 0.0   | 0.0 $\pm$ 0.0             | 0.0 $\pm$ 0.0   | 0.0 $\pm$ 0.0   |
| when-in-rome         | score | 1.0 $\pm$ 0.0            | 1.0 $\pm$ 0.0 | 1.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0           | 0.0 $\pm$ 0.0      | 0.0 $\pm$ 0.0   | 1.0 $\pm$ 0.0              | 1.0 $\pm$ 0.0   | 1.0 $\pm$ 0.0   | 0.0 $\pm$ 0.0             | 0.0 $\pm$ 0.0   | 0.0 $\pm$ 0.0   |
| wikifonia            | score | 1.0 $\pm$ 0.0            | 1.0 $\pm$ 0.0 | 1.0 $\pm$ 0.0  | 0.0 $\pm$ 0.0           | 0.0 $\pm$ 0.0      | 0.0 $\pm$ 0.0   | 1.0 $\pm$ 0.0              | 0.92 $\pm$ 0.17 | 0.75 $\pm$ 0.35 | 0.0 $\pm$ 0.0             | 0.1 $\pm$ 0.2   | 0.11 $\pm$ 0.18 |

**Table 6.** Evaluation of chord and key annotations in the generated JAMS files on the test samples, reported for *times*, *durations*, and *labels* of their observations, and averaged for each partition. Coverage of observation values ranges from 0 (all the expected values are not found in the generated annotation) to 1 (all the expected values are included). Errors are given as seconds (audio) or beats (symbolic) for *times* and *durations*, respectively; and as normalised text similarities for *labels*.

| Partition      | Validated chords | Chord type | Correct conversions | Incorrect conversions | Accuracy $\uparrow$ |
|----------------|------------------|------------|---------------------|-----------------------|---------------------|
| ireal-pro      | 39               | leadsheet  | 37                  | 2                     | 0.949               |
| rock-corpus    | 40               | roman      | 40                  | 0                     | 1.000               |
| weimar         | 37               | leadsheet  | 37                  | 0                     | 1.000               |
| when-in-rome   | 40               | roman      | 40                  | 0                     | 1.000               |
| wikifonia      | 40               | leadsheet  | 39                  | 1                     | 0.975               |
| <i>average</i> | <i>196</i>       | <i>all</i> | <i>193</i>          | <i>3</i>              | <i>0.985</i>        |

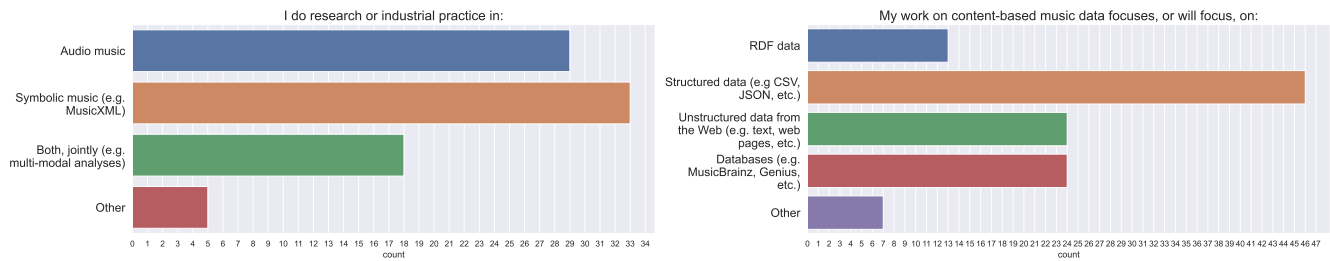
**Table 7.** Evaluation performed by music experts of chord conversion on converted ChoCo partitions.

Also the Semantic Web community would benefit from the introduction of high quality chord data that can be linked to existing Web resources. In turn, this opens up new scenarios and research opportunities for the former communities.

## 5.1 First experiments in Polifonia

An example of novel application at the intersection of both SW and MIR is the Semantic Music Player that the Polifonia consortium demonstrated at the 2021 “AI and Music festival” by SONAR (<https://www.starts.eu/agenda/aimusic-festival-sonar-cccb/detail/>). By leveraging the semantic integration and linking of three collections in ChoCo – including pop (Isophonics), jazz (JAAH), and classical (Schubert Winterreise) music, a mobile app providing an augmented listening experience was developed on top of the resulting KG. During playback of a song, and depending on the user’s preferences and liking, the app can visualise semantic links to related music pieces, depending on controllable musical facets (e.g. harmonic and lyrics similarity) and common entities (e.g. locations, contexts of production). Whereas the harmonic similarity links are enabled by the chordal content of ChoCo, all the other connections were obtained from linking metadata information of these three subsets with MusicBrainz, SecondHandSongs (<https://secondhandsongs.com/>), Songfacts (<https://www.songfacts.com/>), Genius (<https://genius.com/>), and Wikidata. Overall, this application provided an example of how the ChoCo KG can be leveraged for music listening and recommendation, as well as corroborating the potential of SW technologies in such domain – using ontologies to model musical content and relationships explicitly, transparently and meaningfully, as opposed to black-box AI methodologies.

Another line of research that is of particular interest to Polifonia is the creation of music similarity networks<sup>63</sup>, and their consequent investigation through network data analysis techniques<sup>64</sup>. In a music similarity network, nodes typically represent artists, composers, or music pieces (or a relevant grouping of the former), whereas edges express content-specific relationships of similarity. In the context of ChoCo, this has already led to the design of new methods for harmonic similarity, which in turn, are fuelling the creation and the expansion of the *Harmonic Network* (<https://github.com/polifonia-project/lharp>) (induced from the application of these methods on a chord dataset). The computational analysis of the Harmonic Network can indeed reveal interesting insights resulting from the exploration of harmonic relationships from a global perspective,



**Figure 13.** Overview of responses to Questions 2 (music domains, *left*), and 3 (data types, *right*) in the survey.

building upon the local 1-to-1 relationships on which similarity is usually defined. This includes tracing potential influences between authors and across pieces<sup>65</sup>, identifying common harmonic patterns, discovering disruptive artists/pieces<sup>64</sup>, as well as providing analytical support to formulate or test musicological hypotheses. For instance, an algorithmic procedure on the Harmonic Network may discover, or empirically support, that two authors use similar but not identical harmonic structures, even though there is no direct connection between them, but possibly through the influence of a third entity.

## 5.2 Applications and tasks

Given the diversity, size, and quality of the corpus, we expect ChoCo to enable novel applications in Music Technology, other than supporting the design and the evaluation of methods addressing specific tasks in both MIR and computational music analysis. Besides the aforementioned applications in music listening and recommendation, another case study involves the advancement of systems for machine creativity. In the context of our work, these include automatic (or semi-automatic) composition, with particular focus on *arrangement generation*<sup>66</sup> (generating a chord progression, possibly given a melody to accompany); and *melody generation* through harmonic conditioning<sup>10</sup> (generating a melody to play along with a chord progression that is provided as a harmonic template). In ChoCo v1.0, this is enabled by the integrated *Wikifonia* and *Nottingham* collections; and in future versions, with melodic data from *Rock Corpus*, *Weimar*, and the *Band-in-a-Box* collections.

Not only does ChoCo support the creative capabilities of such systems – by providing a considerable amount of quality training data, but it also contributes to their automatic *evaluation*. In fact, the evaluation of music generation systems has recently attracted a growing interest in the field, due to the concerning ethical implications these tools are raising<sup>67</sup>. On one hand, this involves the extraction of statistical features quantifying the degree of alignment between a generated repertoire and the training material, with respect to certain musical properties<sup>68</sup>; on the other hand, it concerns the detection of potential sources of plagiarism in generated music within and beyond the training set<sup>69</sup>.

Another application domain that can benefit from the Chord Corpus is that of *music pedagogy*. For example, TheoryTab (<https://www.hooktheory.com/theorytab>) allows users to choose from a repertoire of popular songs and visualise their harmonic/melodic structure during playback – with chords encoded in both Leadsheet and Roman notations, and projected in such a way as to facilitate the theoretical understanding of the song. Chordify uses chord recognition systems to infer and align chord progressions from audio recordings, and provides support for practising them with guitar, piano, and ukulele. Despite their value, both the technology and the data powering these commercial tools are not openly available, thereby decreasing their overall wider use. In contrast, ChoCo provides an attractive open and linked solution, with its modular architecture enabling the semantic description of chords according to the desired level of complexity and granularity (e.g. an educational ontology for chords might provide a simpler vocabulary). This makes it more suitable for educational purposes.

In the context of MIR, the use of ChoCo as a collection, development/test set, ontology, and knowledge graph, can support a multitude of tasks. The nature of its contribution is twofold: (i) it provides an unprecedented amount of training data, which is often essential for the effectiveness of supervised methods; (ii) it contributes to the development of graph-based methodologies for music analysis that can leverage the semantic representation of chord progressions. For instance, a central research area in MIR is *music similarity*, which in turn encompasses a number of interrelated tasks, including *cover song detection* – useful for music cataloguing and to support court decisions in music plagiarism<sup>70</sup>; and content-based *music retrieval*, aiming to search scores or performances from musical repositories using either symbolic queries, singing (alias query-by-humming), or by playing a smart instrument<sup>71</sup>. Another example of a MIR task that would benefit from ChoCo is *music structure analysis*<sup>72</sup>, which is concerned with the detection and labelling of structural segments related to musical form – a task that strongly relies on the use of harmonic/melodic features<sup>73</sup>. Another task of interest is *music classification* or tagging<sup>74</sup>, a term used to denote a family of more specific problems, including *music genre/style classification*, *composer/artist identification*, and so forth. Finally, examples of tasks of musicological interest that would benefit from ChoCo include *pattern mining*, *cadence detection*, and *local key identification*.

| ChoCo subset                            | Original license | ChoCo license   |
|-----------------------------------------|------------------|-----------------|
| Isophonics                              | Not specified    | CC BY 4.0       |
| JAAH                                    | CC BY-NC-SA 4.0  | CC BY-NC-SA 4.0 |
| Schubert-Winterreise                    | CC BY 3.0        | CC BY 4.0       |
| Billboard                               | CC0              | CC BY 4.0       |
| Chordify Annotator Subjectivity Dataset | CC BY-NC-SA 4.0  | CC BY-NC-SA 4.0 |
| Robbie Williams                         | Not specified    | CC BY 4.0       |
| Uspop-2002                              | Not specified    | CC BY 4.0       |
| RWC-Pop                                 | Not specified    | CC BY 4.0       |
| Real Book                               | Not specified    | CC BY 4.0       |
| Weimar Jazz Database                    | ODbL             | CC BY 4.0       |
| Wikifonia                               | public domain    | CC BY 4.0       |
| iReal Pro                               | public domain    | CC BY 4.0       |
| Band-in-a-box                           | Not specified    | CC BY 4.0       |
| When in Rome                            | CC BY-SA 3.0     | CC BY 4.0       |
| Rock Corpus                             | CC BY 4.0        | CC BY 4.0       |
| Mozart Piano Sonata                     | CC BY-NC-SA 4.0  | CC BY-NC-SA 4.0 |
| Jazz Corpus                             | Not specified    | CC BY 4.0       |
| Nottingham                              | Not specified    | CC BY 4.0       |

**Table 8.** Licensing per ChoCo subset. The second column details the licence declared by the data curator of the corresponding subset; it indicates “*not specified*” whenever this information was not made explicit in articles, web-pages, collection metadata, repositories, etc. The last column refers to the license attributed to the standardisation-integration output for each subset within ChoCo – which is made compliant to the former, as derivative work. Please, note that all the authors of the “*not specified*” subsets were contacted to verify whether the use of a CC-BY license was compliant to their data publishing policies.

### 5.3 Online survey

Since ChoCo is a new resource for the SW, MIR and Musicology communities, we discuss here evidence for potential adoption. To gather such evidence, we performed an online survey in which we directly ask potential adopters 10 questions regarding their background, relevance, and interest in working with chord data. The online survey was distributed in the SW, International Society for MIR (ISMIR), and Digital Musicology mailing lists, gathering a total of  $N = 53$  responses. The survey was conducted via Google Forms – without recording any personal data from participants or any contact information.

Results are illustrated in Figures 13 and 14. Except for questions 1-3 and 12 (multiple choices), all questions ask participants to quantify the agreement with the statement made from 1 (absolutely disagree) to 5 (absolutely agree), 3 being a neutral response (neither agree nor disagree). In the first three questions we assess the background of the respondents, finding that 38 work in MIR, 27 in Musicology, 13 in Semantic Web, and 5 are also involved in other fields (AI, Music Theory, Music Interaction). Most respondents do research or industrial practice using audio (29) or symbolic music (33), or both (18), focusing primarily on structured data when conducting content-based music studies (Figure 13). Nevertheless, music researchers also make extensive use of unstructured data and music databases, and 13 of them (24% of respondents), utilise RDF data.

From questions 4-11 we found that: 64% of respondents have encountered the need for chord datasets providing high-quality timed annotations of harmonic progressions, covering one or more genres/styles; 47% believe that currently existing chord datasets are not of sufficient size for their practise (whereas 41.5% have a neutral position); about 60.3% argue that such datasets do not provide content metadata sufficiently rich and informative to their needs (with another 35.8% being neutral); and 51% believe that links to external resources (e.g. MusicBrainz, Wikidata, etc.) are rarely provided (40% are neutral). Each research community strongly recognises the value of a dataset like ChoCo as a key resource for their field: MIR (91.3%), Semantic Web (71.4%), Musicology (92.7%), and overall, 75.4% of respondents expressed their interest in using such a dataset.

### Code Availability

The ChoCo dataset and Knowledge Graph, together with the ontological ecosystem and code, are publicly available from several repositories (c.f. Table 2). As detailed in Section 2, ChoCo is currently released in 2 modalities:

- As a JAMS dataset, where audio and score annotations are distinguished by the `type` attribute in their `Sandbox`; and temporal/metrical information is expressed in seconds (for audio) and `measure:beat` (for scores) (c.f. Section 2.2);

- As a Knowledge Graph, based on our JAMS ontology to model music annotations (c.f. Section 2.3), and on the Chord and Roman ontologies to semantically describe chords; Table 2 also provides links to a live SPARQL endpoint.

We have implemented a number of actions to ensure that these outputs are in compliance with the FAIR Guiding Principles for scientific data management and stewardship<sup>21</sup>. A GitHub repository hosts data, code, and instructions (<https://github.com/smashub/choco>), to fully reproduce the corpus creation from the original collections. To improve reproducibility, the repository also provides a Docker image for the project (platform agnostic). To improve data consistency, both the latest versions of ChoCo (JAMS file and RDF triples) are available on Zenodo, in synchronisation with GitHub releases.

Via GitHub and Zenodo, the ChoCo project has a unique and persistent identifier and is registered in a searchable source. Additionally, via our integration framework, ChoCo contains fine-grained provenance descriptions that allow to keep track of the original source of each harmonic annotation – both in terms of annotators (the person who contributed the harmonic analysis) and data curator (the maintainer of the original collection).

Finally, to comply with the regulations of the original collections being integrated, all data and code in ChoCo is released under the CC-BY 4.0 license, with the exception of the JAAH, CASD, and Mozart Piano Sonata subsets – which follow the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 international license. This required an in-depth analysis of the licensing policies of the integrated collections (see Table 8). Indeed, for 7 collections, we could not find any specific licensing information from related scientific articles, technical reports, online resources, repositories, dataset metadata, and so forth. For these cases, the authors of these collections were contacted to investigate whether the use of the CC-BY 4.0 license – on our derivative integration work – was compatible with their original releasing strategies.

## References

1. Cook, N. D. & Fujisawa, T. X. The psychophysics of harmony perception: Harmony is a three-tone phenomenon. *Empir. Music. Rev.* (2006).
2. Chan, P. Y., Dong, M. & Li, H. The science of harmony: A psychophysical basis for perceptual tensions and resolutions in music. *Research* (2019).
3. Grove, G., Sadie, S. & Stolba, K. M. *The New Grove Dictionary of Music and Musicians* (Macmillan Publishers, London ; Washington, D.C., 1980).
4. Piston, W. *Harmony* (W. W. Norton, 1962).
5. Schoenberg, A., Carter, R. & Frisch, W. *Theory of Harmony* (University of California, 2010).
6. Hentschel, J., Neuwirth, M. & Rohrmeier, M. The annotated Mozart Sonatas: Score, Harmony, and Cadence. *Transactions Int. Soc. for Music. Inf. Retr.* **4** (2021).
7. Koops, H. V. *et al.* Annotator subjectivity in harmony annotations of popular music. *J. New Music. Res.* **48**, 232–252 (2019).
8. Hindemith, P. & Mendel, A. *The craft of musical composition. 1. Theoretical part* (Schott, 1970).
9. McLachlan, N., Marco, D., Light, M. & Wilson, S. Consonance and pitch. *J. Exp. Psychol. Gen.* **142**, 1142 (2013).
10. Eck, D. & Schmidhuber, J. A first look at music composition using LSTM recurrent neural networks. *Istituto Dalle Molle Di Studi Sull Intelligenza Artif.* **103**, 48 (2002).
11. Carriero, V. A. *et al.* Semantic integration of MIR datasets with the polifonia ontology network. In *Proc. of the International Society for Music Information Retrieval Conference, ISMIR* (2021).
12. Hentschel, J., Moss, F. C., McLeod, A., Neuwirth, M. & Rohrmeier, M. Towards a Unified Model of Chords in Western Harmony. In Münnich, S. & Rizo, D. (eds.) *Music Encoding Conference Proceedings 2021*, 143–149 (Humanities Commons, 2022).
13. Humphrey, E. J. *et al.* JAMS: A JSON Annotated Music Specification for Reproducible MIR Research. In *International Conference on Music Information Retrieval (ISMIR)*, 591–596 (2014).
14. Bittner, R. *et al.* mirdata: Software for Reproducible Usage of Datasets. In *Proceedings of the 20th International Society for Music Information Retrieval Conference*, 99–106, [10.5281/zenodo.3527750](https://doi.org/10.5281/zenodo.3527750) (International Conference on Music Information Retrieval (ISMIR), Delft, The Netherlands, 2019).
15. Raffel, C. *et al.* MIR\_EVAL: A Transparent Implementation of Common MIR Metrics. In *International Conference on Music Information Retrieval (ISMIR)*, 367–372 (2014).
16. Hogan, A. *et al.* Knowledge graphs. *Synth. Lect. on Data, Semantics, Knowl.* **12**, 1–257 (2021).



17. De Haas, W. B., Magalhaes, J. P., Ten Heggeler, D., Bekenkamp, G. & Ruizendaal, T. Chordify: Chord transcription for the masses. In *Demonstration presented at the International Society for Music Information Retrieval Conference*, 8–12 (2012).
18. Weigl, D. M. *et al.* Fair interconnection and enrichment of public-domain music resources on the web. *Empir. Music. Rev.* **16**, 16–33 (2021).
19. Gotham, M. R. Connecting the dots: Engaging wider forms of openness for the mutual benefit of musicians and musicologists. *Empir. Music. Rev.* **16**, 34–46 (2021).
20. Jensenius, A. R. Best versus good enough practices for open music research. *Empir. Music. Rev.* **16**, 5–15 (2021).
21. Wilkinson, M. D. *et al.* The FAIR Guiding Principles for scientific data management and stewardship. *Sci. data* **3**, 1–9 (2016).
22. Raimond, Y., Abdallah, S., Sandler, M. & Giasson, F. The Music Ontology. In *International Conference on Music Information Retrieval (ISMIR)* (2007).
23. Sutton, C., Raimond, Y. & Mauch, M. The OMRAS2 Chord Ontology. <http://purl.org/ontology/chord/> (2007).
24. Poltronieri, A. & Gangemi, A. The hamse ontology: Using semantic technologies to support music representation interoperability and musicological analysis. In *Proceedings of the 1st Workshop on Multisensory Data and Knowledge (MDK 2021)* (2021).
25. Jones, J., de Siqueira Braga, D., Tertuliano, K. & Kauppinen, T. Musicowl: the Music Score Ontology. In *Proceedings of the International Conference on Web Intelligence*, 1222–1229 (2017).
26. Rashid, S. M., De Roure, D. & McGuinness, D. L. A Music Theory Ontology. In *Proceedings of the 1st International Workshop on Semantic Applications for Audio and Music*, 6–14 (2018).
27. Guillotel-Nothmann, C. & Ceulemans, A.-E. Das diatonisch-chromatische System zur Zeit des Michael Praetorius. Eine digitale Neuerschließung des Syntagma Musicum (1619) in Verbindung mit dem Tanzzyklus Terpsichore (1612). *Musik im Umbruch. Michael Praetorius zum 400* (2022).
28. Meroño-Peñuela, A. *et al.* The MIDI linked data cloud. In *International Semantic Web Conference*, 156–164 (Springer, 2017).
29. de Berardinis, J., Meroño-Peñuela, A., Poltronieri, A. & Presutti, V. ChoCo: the Chord Corpus, [10.5281/zenodo.7193888](https://doi.org/10.5281/zenodo.7193888) (2023).
30. Harte, C., Sandler, M. B., Abdallah, S. A. & Gómez, E. Symbolic Representation of Musical Chords: A Proposed Syntax for Text Annotations. In *International Conference on Music Information Retrieval (ISMIR)*, vol. 5, 66–71 (2005).
31. Mauch, M. *et al.* OMRAS2 metadata project 2009. In *12th International Society for Music Information Retrieval Conference, ISMIR* (2009).
32. Eremenko, V., Demirel, E., Bozkurt, B. & Serra, X. JAAH: Audio-aligned jazz harmony dataset, [10.5281/zenodo.1290737](https://doi.org/10.5281/zenodo.1290737) (2018).
33. Weiß, C. *et al.* Schubert Winterreise dataset: A multimodal scenario for music analysis. *J. on Comput. Cult. Herit. (JOCCH)* **14**, 1–18 (2021).
34. Burgoyne, J. A., Wild, J. & Fujinaga, I. An Expert Ground Truth Set for Audio Chord Recognition and Music Analysis. In *ISMIR*, vol. 11, 633–638 (2011).
35. Di Giorgi, B., Zanoni, M., Sarti, A. & Tubaro, S. Automatic chord recognition based on the probabilistic modeling of diatonic modal harmony. In *Proceedings of the 8th International Workshop on Multidimensional Systems*, 1–6 (VDE, 2013).
36. Mauch, M., Dixon, S., Harte, C., Casey, M. A. & Fields, B. Discovering Chord Idioms Through Beatles and Real Book Songs. In Dixon, S., Bainbridge, D. & Typke, R. (eds.) *Proceedings of the 8th International Conference on Music Information Retrieval, ISMIR 2007, Vienna, Austria, September 23-27, 2007*, 255–258 (Austrian Computer Society, 2007).
37. Berenzweig, A., Logan, B., Ellis, D. P. & Whitman, B. A large-scale evaluation of acoustic and subjective music-similarity measures. *Comput. Music. J.* 63–76 (2004).
38. Goto, M., Hashiguchi, H., Nishimura, T. & Oka, R. RWC Music Database: Popular, Classical and Jazz Music Databases. In *Ismir*, vol. 2, 287–288 (2002).
39. Pfeleiderer, M., Frieler, K., Abeßer, J., Zaddach, W.-G. & Burkhart, B. (eds.) *Inside the Jazzomat - New Perspectives for Jazz Research* (Schott Campus, 2017).

- 752 **40.** De Haas, W. B., Robine, M., Hanna, P., Veltkamp, R. C. & Wiering, F. Comparing approaches to the similarity of musical  
753 chord sequences. In *International Symposium on Computer Music Modeling and Retrieval*, 242–258 (Springer, 2010).
- 754 **41.** Micchi, G., Gotham, M. & Giraud, M. Not all roads lead to Rome: Pitch representation and model architecture for  
755 automatic harmonic analysis. *Transactions Int. Soc. for Music. Inf. Retr. (TISMIR)* **3**, 42–54 (2020).
- 756 **42.** De Clercq, T. & Temperley, D. A corpus analysis of rock harmony. *Pop. Music.* **30**, 47–70 (2011).
- 757 **43.** Granroth-Wilding, M. & Steedman, M. A robust parser-interpreter for jazz chord sequences. *J. New Music. Res.* **43**,  
758 355–374 (2014).
- 759 **44.** Nottingham Database. <https://ifdo.ca/~seymour/nottingham/nottingham.html>. Accessed: 2022-05-05.
- 760 **45.** *The Real Book* (Hal Leonard Publishing Corporation, 2004).
- 761 **46.** Feist, J. *Berklee Contemporary Music Notation* (Hal Leonard Corporation, 2017).
- 762 **47.** Andrews, W. G. & Sclater, M. *Materials of western music* (Alfred Music Publishing, 1997).
- 763 **48.** Devaney, J., Arthur, C., Condit-Schultz, N. & Nisula, K. Theme and variation encodings with roman numerals (TAVERN):  
764 A new data set for symbolic music analysis. In *International Society for Music Information Retrieval Conference (ISMIR)*  
765 (2015).
- 766 **49.** Chen, T.-P., Su, L. *et al.* Functional Harmony Recognition of Symbolic Music Data with Multi-task Recurrent Neural  
767 Networks. In *ISMIR*, 90–97 (2018).
- 768 **50.** Yeh, Y.-C. *et al.* Automatic melody harmonization with triad chords: A comparative study. *J. New Music. Res.* **50**, 37–51  
769 (2021).
- 770 **51.** Xi, Q., Bittner, R. M., Pauwels, J., Ye, X. & Bello, J. P. GuitarSet: A Dataset for Guitar Transcription. In *ISMIR*, 453–460  
771 (2018).
- 772 **52.** Barbancho, A. M., Barbancho, I., Tardón, L. J. & Molina, E. *Database of Piano Chords: An Engineering View of Harmony*  
773 (Springer, 2013).
- 774 **53.** Wang, Z. *et al.* POP909: A Pop-song Dataset for Music Arrangement Generation. In *Proceedings of 21st International*  
775 *Conference on Music Information Retrieval, ISMIR* (2020).
- 776 **54.** Temperley, D. Nottingham Database. <http://davidtemperley.com/kp-stats/>. Accessed: 2022-05-05.
- 777 **55.** Gebru, T. *et al.* Datasheets for datasets. *Commun. ACM* **64**, 86–92 (2021).
- 778 **56.** Cuthbert, M. S. & Ariza, C. music21 : A Toolkit for Computer-Aided Musicology and Symbolic Music Data. In  
779 *International Society for Music Information Retrieval Conference (ISMIR)*, Ismir, 637–642 (2010).
- 780 **57.** de Berardinis, J., Meroño-Peñuela, A., Poltronieri, A. & Presutti, V. The Music Annotation Pattern. In Svátek, V., Carriero,  
781 V. A., Poveda, M., Kindermann, C. & Zhou, L. (eds.) *Proceedings of the 13th Workshop on Ontology Design and Patterns*  
782 *(WOP 2023)* (2022).
- 783 **58.** Gangemi, A. Ontology Design Patterns for Semantic Web Content. In Gil, Y., Motta, E., Benjamins, V. R. & Musen,  
784 M. A. (eds.) *The Semantic Web - ISWC 2005, 4th International Semantic Web Conference, ISWC 2005, Galway, Ireland,*  
785 *November 6-10, 2005, Proceedings*, vol. 3729 of *Lecture Notes in Computer Science*, 262–276, [10.1007/11574620\\_21](https://doi.org/10.1007/11574620_21)  
786 (Springer, 2005).
- 787 **59.** Presutti, V., Daga, E., Gangemi, A. & Blomqvist, E. eXtreme Design with Content Ontology Design Patterns. In  
788 Blomqvist, E., Sandkuhl, K., Scharffe, F. & Svátek, V. (eds.) *Proceedings of the Workshop on Ontology Patterns (WOP*  
789 *2009)*, *collocated with the 8th International Semantic Web Conference (ISWC)*, vol. 516 of *CEUR Workshop Proceedings*  
790 (CEUR-WS.org, 2009).
- 791 **60.** Lebo, T. *et al.* Prov-o: The prov ontology (2013).
- 792 **61.** Daga, E., Asprino, L., Mulholland, P. & Gangemi, A. Facade-X: An Opinionated Approach to SPARQL Anything. In  
793 Alam, M., Groth, P., de Boer, V., Pellegrini, T. & Pandit, H. J. (eds.) *Volume 53: Further with Knowledge Graphs*, vol. 53,  
794 58–73 (IOS Press, 2021).
- 795 **62.** Adamou, A., Brown, S., Barlow, H., Allocca, C. & d’Aquin, M. Crowdsourcing Linked Data on listening experiences  
796 through reuse and enhancement of library data. *Int. J. on Digit. Libr.* **20**, 61–79 (2019).
- 797 **63.** Ono, J., Corrêa, D., Ferreira, M., Mello, R. & Nonato, L. G. Similarity graph: Visual exploration of song collections. In  
798 *SIBGRAPI* (IEEE, Institute of Electrical and Electronics Engineers United States, 2015).

- 799 **64.** Andrade, F. F. N., Figueiredo, F., Silva, D. & Morais, F. Measuring disruption in song similarity networks. In *International*  
800 *Conference on Music Information Retrieval (ISMIR)* (2020).
- 801 **65.** Morton, B. G. & Kim, Y. E. Acoustic features for recognizing musical artist influence. In *2015 IEEE 14th International*  
802 *Conference on Machine Learning and Applications (ICMLA)*, 1117–1122 (IEEE, 2015).
- 803 **66.** Simon, I., Morris, D. & Basu, S. MySong: automatic accompaniment generation for vocal melodies. In *Proceedings of the*  
804 *SIGCHI conference on human factors in computing systems*, 725–734 (2008).
- 805 **67.** Sturm, B. L., Iglesias, M., Ben-Tal, O., Miron, M. & Gómez, E. Artificial intelligence and music: open questions of  
806 copyright law and engineering praxis. In *Arts*, vol. 8, 115 (MDPI, 2019).
- 807 **68.** Yang, L.-C. & Lerch, A. On the evaluation of generative models in music. *Neural Comput. Appl.* **32**, 4773–4784 (2020).
- 808 **69.** Yin, Z., Reuben, F., Stepney, S. & Collins, T. Measuring When a Music Generation Algorithm Copies Too Much: The  
809 Originality Report, Cardinality Score, and Symbolic Fingerprinting by Geometric Hashing. *SN Comput. Sci.* **3**, 1–18  
810 (2022).
- 811 **70.** Müllensiefen, D. & Pendzich, M. Court decisions on music plagiarism and the predictive value of similarity algorithms.  
812 *Music. Sci.* **13**, 257–295 (2009).
- 813 **71.** Turchet, L., Pauwels, J., Fischione, C. & Fazekas, G. Cloud-smart musical instrument interactions: Querying a large music  
814 collection with a smart guitar. *ACM Transactions on Internet Things* **1**, 1–29 (2020).
- 815 **72.** de Berardinis, J., Vamvakaris, M., Cangelosi, A. & Coutinho, E. Unveiling the hierarchical structure of music by  
816 multi-resolution community detection. *Transactions Int. Soc. for Music. Inf. Retr.* **3**, 82–97 (2020).
- 817 **73.** Bartsch, M. A. & Wakefield, G. H. Audio thumbnailing of popular music using chroma-based representations. *IEEE*  
818 *Transactions on multimedia* **7**, 96–104 (2005).
- 819 **74.** Bertin-Mahieux, T., Eck, D. & Mandel, M. Automatic tagging of audio: The state-of-the-art. In *Machine audition:*  
820 *Principles, algorithms and systems*, 334–352 (IGI Global, 2011).

## 821 Acknowledgements

822 This project has received funding from the European Union’s Horizon 2020 research and innovation programme under grant  
823 agreement No 101004746. The authors acknowledge Nicholas Lazzari for his contribution to the data conversion module and  
824 the test JAMS dataset, and Elia Rizzetto for his contribution to the latter. We also thank Simon Holland and Naomi Barker for  
825 their role in the expert validation of the chord conversion module, and all the annotators that took part in this study. The authors  
826 also thank Andrew Choi for his contribution to the Band-in-a-Box parser, Mark Gotham for addressing queries on the When in  
827 Rome corpus, Mark Granroth-Wilding for having provided details of the Jazz Corpus, Luigi Aspirino and Enrico Daga for the  
828 technical support with SPARQL-Anything, and Marilena Daquino for contributing to the deployment of the SPARQL endpoint.

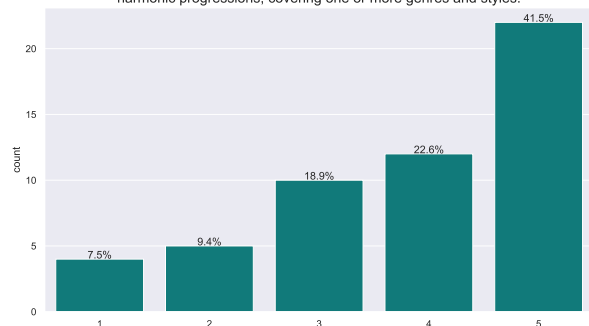
## 829 Author contributions

830 J.d.B. designed the data transformation pipeline of ChoCo, implemented the JAMifier module, carried out the evaluation  
831 studies and the survey, and contributed to the corpus analysis and testing. A.P. designed the JAMS and the RomanChord  
832 ontologies, implemented the Chconverter and jams2rdf modules, carried out the evaluation studies, and deployed the  
833 SPARQL endpoint. A.M. contributed to the jams2rdf module, link discovery, and coordinated the survey. V.P. designed and  
834 tested the ontological ecosystem of ChoCo. All authors have read and approved the final manuscript.

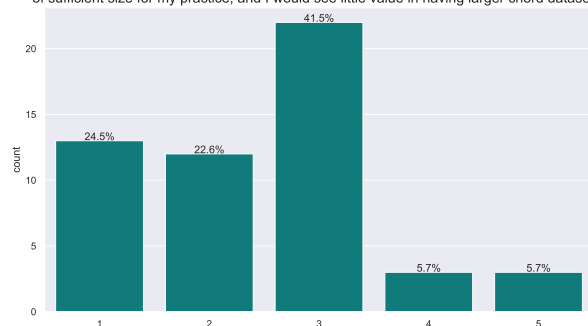
## 835 Competing interests

836 The author(s) declare no competing interests.

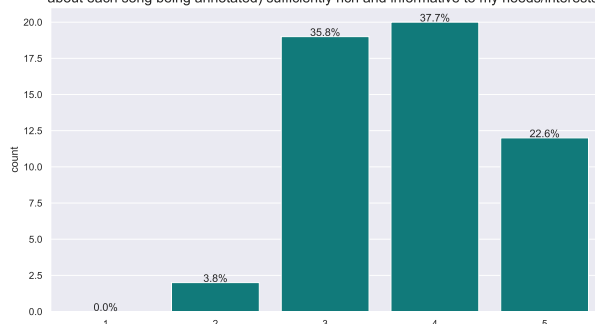
I have encountered the need for chord datasets providing high-quality and timed annotations of harmonic progressions, covering one or more genres and styles.



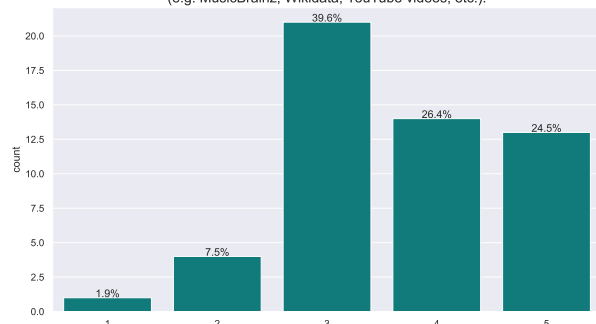
From my experience, currently existing chord datasets (with high-quality and timed annotations) are of sufficient size for my practice, and I would see little value in having larger chord datasets.



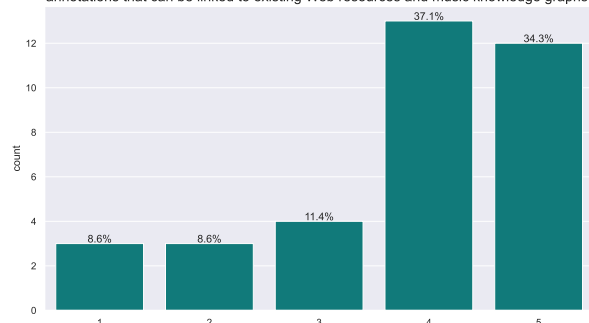
From my experience, currently existing chord datasets do not provide content metadata (information about each song being annotated) sufficiently rich and informative to my needs/interests.



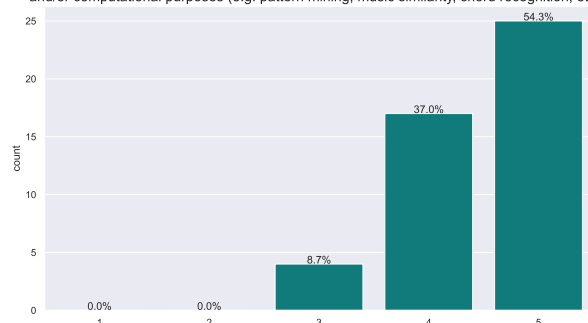
From my experience, currently existing chord datasets rarely provide links to external resources (e.g. MusicBrainz, Wikidata, YouTube videos, etc.).



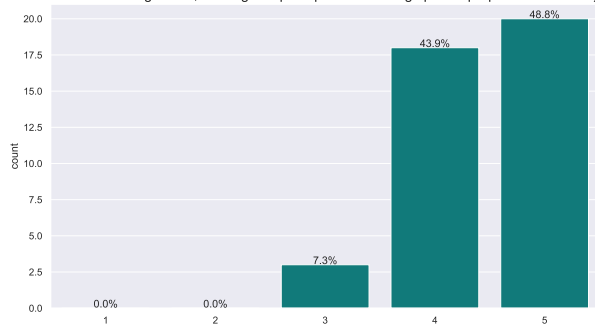
I think that the Semantic Web community would benefit from a large collection of quality chord annotations that can be linked to existing Web resources and music knowledge graphs.



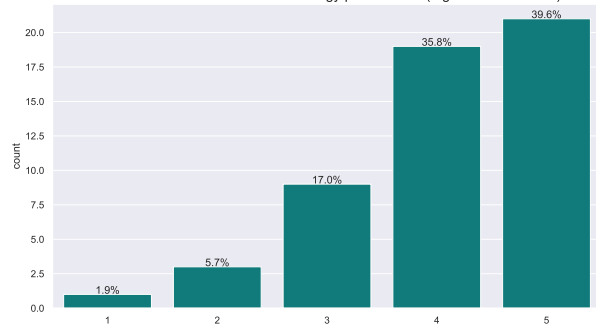
I think that the MIR community would benefit from a large collection of diverse, quality chord annotations, consistently aligned to/across different notations, which can be used for analytical and/or computational purposes (e.g. pattern mining, music similarity, chord recognition, etc.).



I think that the Musicology community would benefit from a large collection of diverse, quality chord annotations that can be used for analytical purposes and large scale studies (e.g. tracing influence among artists, running complex queries entailing specific properties of chords).



Overall, I would be interested in using a large corpus of diverse, timed, and quality chord annotations that is released in such a way as to accommodate both the interests and needs of Semantic Web and/or Music Technology practitioners (e.g. JAMS and RDF).



**Figure 14.** Questions and overview of responses for Questions 4-11 from the online survey.